

THE BLOW-UP FORMULA IN PRISMATIC COHOMOLOGY WITH NYGAARD COMPATIBILITY

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ABSTRACT. Let X be a regular scheme, proper and smooth over $\mathcal{O}_K$, and let $i: Z \hookrightarrow X$ be a regular closed immersion of codimension c . We prove that the prismatic complex of the blow-up $\tilde{X} = \mathrm{Bl}_Z X$ decomposes canonically as

$$R\Gamma_{\mathrm{Prism}}(\tilde{X}) \simeq R\Gamma_{\mathrm{Prism}}(X) \oplus \bigoplus_{r=1}^{c-1} R\Gamma_{\mathrm{Prism}}(Z)(-r)[-2r]$$

in the prismatic derived category $D_{\mathrm{pris}}^{\varphi, N}(A_{\mathrm{inf}})$. The isomorphism is canonical, φ -equivariant, and compatible with the Nygaard filtration $N^\bullet$.

This result is expected by experts as a consequence of the formalism of Bhatt–Morrow–Scholze and Bhatt–Scholze, but to our knowledge has not appeared in the literature with the Nygaard compatibility made explicit and verified. We give a complete proof with full details on: the rational isomorphism via faithful flatness arguments; the integral descent via the Deligne–Illusie criterion; the Nygaard compatibility verified map-by-map; and explicit torsion bounds.

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1. INTRODUCTION

1.1. Setup and statement. Let p be a prime, K a complete p -adic field with ring of integers $\mathcal{O}_K$ and perfect residue field k . Write $W(k)$ for the Witt vectors of k and $A_{\mathrm{inf}} = W(\mathcal{O}_C^b)$ for Fontaine’s period ring, where C is a completed algebraic closure of K .

We write $R\Gamma_{\text{Prism}}(X)$ for the prismatic complex of a p -adic formal scheme X , as constructed in [1, 2], and $D_{\text{pris}}^{\varphi, N}(A_{\text{inf}})$ for the derived category of perfect A_{inf} -complexes equipped with a semilinear Frobenius φ and a descending Nygaard filtration $N^\bullet$.

Theorem 1.1 (Main theorem). *Let X be regular, proper and smooth over $\mathcal{O}_K$, and $i: Z \hookrightarrow X$ a regular closed immersion of codimension $c \geq 1$. Let $\pi: \tilde{X} = \text{Bl}_Z X \rightarrow X$ be the blow-up along Z , with exceptional divisor $E = \pi^{-1}(Z) \simeq \mathbb{P}(N_{Z/X})$.*

(1) (Rational.) *There is a canonical isomorphism in $D_{\text{pris}}^{\varphi, N}(A_{\text{inf}}[1/p])$:*

$$(1) \quad R\Gamma_{\text{Prism}}(\tilde{X}) \simeq R\Gamma_{\text{Prism}}(X) \oplus \bigoplus_{r=1}^{c-1} R\Gamma_{\text{Prism}}(Z)(-r)[-2r],$$

φ -equivariant and compatible with $N^\bullet$.

(2) (Integral.) *Under hypotheses (A)+(B) of Proposition 6.1, the isomorphism (1) holds in $D_{\text{pris}}^{\varphi, N}(A_{\text{inf}})$.*

1.2. Precise inputs from [1, 2]. We list the results from the foundational papers that enter the proof formally, to distinguish them from analogies.

- (1) [1, Theorem 1.8]: perfectness of $R\Gamma_{\text{Prism}}(X)$, finiteness of p -torsion, and the comparisons with A_{cris} and B_{dR}^+ .
- (2) [1, Prop. 4.13]: the p -torsion $H_{\text{Prism}}^i(X)[p^\infty]$ is finite.
- (3) [1, Theorem 1.8(4)]: the Hodge–Tate comparison $\text{gr}_N^j H_{\text{Prism}}^i(X) \otimes_{A_{\text{inf}}, \theta} W(k) \simeq H^{i-j}(X, \Omega_{X/\mathcal{O}_K}^j)$.
- (4) [2, Section 7]: construction of prismatic Chern classes $c_r^{\text{Prism}}(\mathcal{E})$, their φ -equivariance, and $c_1^{\text{Prism}}(\mathcal{L}) \in N^1 H_{\text{Prism}}^2(X)(1)$.
- (5) [2, Section 9]: purity/Gysin isomorphism $R\Gamma_{\text{Prism}, Z}(X) \simeq R\Gamma_{\text{Prism}}(Z)(-c)[-2c]$ for $Z \hookrightarrow X$ a regular closed immersion of codimension c (the essential input for Lemma 3.2).
- (6) [1, Theorem 5.2]: proper base change for the pushforward $R\pi_*$.

1.3. What is new. The formula (1) follows formally from (1)–(6) above, via the classical strategy of Illusie [5] transported to the prismatic setting. A specialist familiar with [1, 2] would consider it expected. Our contributions are:

- (i) *Nygaard compatibility, map by map.* We track the Nygaard filtration explicitly through the projective bundle formula, the Gysin isomorphism, and the splitting argument, verifying at each step that the maps respect the filtration (Section 5).
- (ii) *Rational $\rightarrow$ integral passage.* We give a complete argument for the integral version using the Deligne–Illusie criterion [4], with an explicit discussion of the extension obstructions that must vanish (Proposition 6.1).
- (iii) *Effective torsion bounds.* We give explicit bounds on the p -torsion exponent of $H_{\text{Prism}}^i(\tilde{X})$ in terms of the Hodge cohomology of X and Z (Proposition 6.2).

We emphasize that this is a foundational result in prismatic geometry and does not imply any of the major open conjectures.

2. PRELIMINARIES

2.1. Prismatic cohomology: the facts we use. For X proper smooth over $\mathcal{O}_K$, the prismatic complex $R\Gamma_{\text{Prism}}(X) \in D_{\text{pris}}^{\varphi, N}(A_{\text{inf}})$ satisfies:

- (1) *Perfectness.* Each $H_{\text{Prism}}^i(X)$ is a finitely generated, p -adically complete A_{inf} -module, and $R\Gamma_{\text{Prism}}(X)$ is perfect as an A_{inf} -complex [1, Theorem 1.8].
- (2) *Finite torsion.* The p -torsion $H_{\text{Prism}}^i(X)[p^\infty]$ is finite [1, Prop. 4.13]. Explicitly, there exists $N = N(X, i) \geq 0$ such that $p^N \cdot H_{\text{Prism}}^i(X)[p^\infty] = 0$.

- (3) *Comparison with crystalline.* There is a canonical φ -equivariant isomorphism [1, Theorem 1.8(ii)]:

$$\alpha_{A_{\text{cris}}}(X): R\Gamma_{\text{Prism}}(X) \otimes_{A_{\text{inf}}}^L A_{\text{cris}} \xrightarrow{\sim} R\Gamma_{\text{cris}}(X/W(k)).$$

- (4) *Comparison with de Rham.* There is a canonical φ -equivariant isomorphism [1, Theorem 1.8(iii)]:

$$\alpha_{B_{\text{dR}}^+}(X): R\Gamma_{\text{Prism}}(X) \otimes_{A_{\text{inf}}}^L B_{\text{dR}}^+ \xrightarrow{\sim} R\Gamma_{\text{dR}}(X_K),$$

with the Nygaard filtration corresponding to the Hodge filtration.

- (5) *Hodge–Tate graded pieces.* For each $j \geq 0$, the comparison [1, Theorem 1.8(4)]:

$$\text{gr}_N^j H_{\text{Prism}}^i(X) \otimes_{A_{\text{inf}}, \theta} W(k) \simeq H^{i-j}(X, \Omega_{X/\mathcal{O}_K}^j),$$

where $\theta: A_{\text{inf}} \rightarrow W(k)$ is the natural map.

2.2. Prismatic cohomology with support. For a closed immersion $i: Z \hookrightarrow X$ with complementary open $j: U = X \setminus Z$, define:

$$R\Gamma_{\text{Prism}, Z}(X) := \text{Cone}(R\Gamma_{\text{Prism}}(X) \xrightarrow{j^*} R\Gamma_{\text{Prism}}(U))[-1].$$

Remark 2.1. The map j^* is φ -equivariant and $N^\bullet$ -compatible (because the restriction functor j^* is a morphism in $D_{\text{pris}}^{\varphi, N}$). Therefore $R\Gamma_{\text{Prism}, Z}(X)$ inherits a φ -action and a Nygaard filtration:

$$N^s R\Gamma_{\text{Prism}, Z}(X) := \text{Cone}(N^s R\Gamma_{\text{Prism}}(X) \xrightarrow{j^*} N^s R\Gamma_{\text{Prism}}(U))[-1],$$

and the diagram

$$\begin{array}{ccccc} N^s R\Gamma_{\text{Prism}, Z}(X) & \rightarrow & N^s R\Gamma_{\text{Prism}}(X) & \rightarrow & N^s R\Gamma_{\text{Prism}}(U) \\ \downarrow & & \downarrow & & \downarrow \\ R\Gamma_{\text{Prism}, Z}(X) & \rightarrow & R\Gamma_{\text{Prism}}(X) & \rightarrow & R\Gamma_{\text{Prism}}(U) \end{array}$$

commutes in $D(A_{\text{inf}})$. This is the Nygaard-filtered excision triangle.

2.3. Prismatic Chern classes. The prismatic first Chern class $c_1^{\text{Prism}}(\mathcal{L}) \in H_{\text{Prism}}^2(X)(1)$ for a line bundle $\mathcal{L}$ is constructed in [2, Section 7]. Key properties we use:

- (a) $\varphi(c_1^{\text{Prism}}(\mathcal{L})) = p \cdot c_1^{\text{Prism}}(\mathcal{L})$ (from the Tate twist (1));
- (b) $c_1^{\text{Prism}}(\mathcal{L}) \in N^1 H_{\text{Prism}}^2(X)(1)$ [1, Theorem 1.8(4)];
- (c) under $\alpha_{A_{\text{cris}}}$, $c_1^{\text{Prism}}(\mathcal{L})$ maps to the crystalline Chern class $c_1^{\text{cris}}(\mathcal{L})$ [2, Section 7];
- (d) the cup product with $c_1^{\text{Prism}}(\mathcal{L})$ respects both φ and $N^\bullet$.

3. TWO KEY LEMMAS

3.1. Projective bundle formula.

Lemma 3.1 (Projective bundle formula). *Let $q: \mathbb{P} = \mathbb{P}(\mathcal{E}) \rightarrow Z$ be the projective bundle of a locally free sheaf $\mathcal{E}$ of rank c on Z . Set $\xi = c_1^{\text{Prism}}(\mathcal{O}_{\mathbb{P}}(1)) \in H_{\text{Prism}}^2(\mathbb{P})(1)$. Then:*

$$(2) \quad \Phi: \bigoplus_{r=0}^{c-1} R\Gamma_{\text{Prism}}(Z)(-r)[-2r] \xrightarrow{\sim} R\Gamma_{\text{Prism}}(\mathbb{P}), \quad (\alpha_r) \mapsto \sum_{r=0}^{c-1} q^*(\alpha_r) \cup \xi^r,$$

is an isomorphism in $D_{\text{pris}}^{\varphi, N}(A_{\text{inf}})$.

Proof. We verify three things: Φ is a rational isomorphism, it is an integral isomorphism, and it is φ -equivariant and $N^\bullet$ -compatible.

(a) Rational isomorphism.

The projective bundle formula in crystalline cohomology,

$$\bigoplus_{r=0}^{c-1} R\Gamma_{\text{cris}}(Z)(-r)[-2r] \xrightarrow{\sim} R\Gamma_{\text{cris}}(\mathbb{P}),$$

is proved in [5, Exposé VII, Prop. 3.3]. We apply $(-) \otimes_{A_{\text{inf}}} A_{\text{cris}}$ to Φ . By property (3) of Section 2, the source becomes $\bigoplus_r R\Gamma_{\text{cris}}(Z)(-r)[-2r]$, and the target becomes $R\Gamma_{\text{cris}}(\mathbb{P})$. Moreover, by property (c) of Section 2, $c_1^{\text{Prism}}(\mathcal{O}(1)) \otimes_{A_{\text{inf}}} 1$ maps to $c_1^{\text{cris}}(\mathcal{O}(1))$. Hence $\Phi \otimes_{A_{\text{inf}}} A_{\text{cris}}$ coincides with the classical Illusie isomorphism, which is an isomorphism.

Since $A_{\text{inf}}[1/p] \rightarrow A_{\text{cris}}[1/p]$ is faithfully flat (see [1, Lemma 3.23]), an isomorphism after tensoring with $A_{\text{cris}}[1/p]$ implies an isomorphism after tensoring with $A_{\text{inf}}[1/p]$. Therefore $\Phi \otimes_{A_{\text{inf}}} A_{\text{inf}}[1/p]$ is an isomorphism, i.e., Φ is an isomorphism in $D_{\text{pris}}^{\varphi, N}(A_{\text{inf}}[1/p])$.

(b) Integral isomorphism.

Both source and target are perfect complexes of A_{inf} -modules (property (1) of Section 2). For a map $f: P \rightarrow Q$ between perfect complexes of A_{inf} -modules, f is an isomorphism in $D(A_{\text{inf}})$ if and only if both $f[1/p]$ (an isomorphism by (a)) and the induced map on p -torsion f_{p^∞} are isomorphisms.

We verify f_{p^∞} is an isomorphism by showing both sides have the same p -torsion. By property (2) of Section 2, $H_{\text{Prism}}^i(\mathbb{P})[p^\infty]$ and $\bigoplus_r H_{\text{Prism}}^{i-2r}(Z)[p^\infty]$ are both controlled by the Hodge–Tate graded pieces $H^{i-j}(\mathbb{P}, \Omega^j)$ and $H^{i-2r-j}(Z, \Omega^j)$ respectively, via the spectral sequence of property (5). The classical identification of Hodge cohomology of $\mathbb{P}(\mathcal{E})$ in terms of that of Z (the projective bundle formula for $\Omega^\bullet$, see [5, Exposé VII]) shows that these are equal. Hence Φ_{p^∞} is an isomorphism, and Φ is integral.

(c) φ -equivariance.

On the source, φ acts on the summand $R\Gamma_{\text{Prism}}(Z)(-r)[-2r]$ as p^r times the Frobenius of $R\Gamma_{\text{Prism}}(Z)$ (from the Tate twist $(-r)$). On the target, φ acts naturally on $R\Gamma_{\text{Prism}}(\mathbb{P})$. The map $\Phi(\alpha_r) = q^*(\alpha_r) \cup \xi^r$ is φ -equivariant because:

- q^* is a ring map and hence φ -equivariant;
- $\varphi(\xi^r) = \varphi(c_1^{\text{Prism}}(\mathcal{O}(1)))^r = (p \cdot c_1^{\text{Prism}}(\mathcal{O}(1)))^r = p^r \xi^r$ (property (a) of Section 2);
- so $\varphi(q^*(\alpha_r) \cup \xi^r) = q^*(\varphi(\alpha_r)) \cup p^r \xi^r = p^r \cdot q^*(\varphi(\alpha_r)) \cup \xi^r$, consistent with φ acting by p^r on the r -th summand.

(d) Nygaard compatibility.

We must check that for each $s \geq 0$, the isomorphism Φ restricts to an isomorphism on the s -th Nygaard filtration pieces.

On the source, N^s of the r -th summand $R\Gamma_{\text{Prism}}(Z)(-r)[-2r]$ is $N^{s-r} R\Gamma_{\text{Prism}}(Z)$ (the Tate twist shifts the filtration: a section α of the r -th twist lies in N^s iff $\varphi(\alpha) \in p^s(-r)$ iff $p^r \varphi(\alpha_0) \in p^s(-r)$ for the untwisted class α_0 , iff $\varphi(\alpha_0) \in p^{s-r}$, iff $\alpha_0 \in N^{s-r}$).

On the target, $N^s R\Gamma_{\text{Prism}}(\mathbb{P})$ is defined by the same condition for $\mathbb{P}$.

We need: the class $q^*(\alpha_r) \cup \xi^r$ lies in $N^s R\Gamma_{\text{Prism}}(\mathbb{P})$ iff $\alpha_r \in N^{s-r} R\Gamma_{\text{Prism}}(Z)$.

For this, note:

- $\xi = c_1^{\text{Prism}}(\mathcal{O}(1)) \in N^1$ (property (b) of Section 2), so $\xi^r \in N^r$ (by multiplicativity of the cup product with the filtration: if $a \in N^{r_1}$ and $b \in N^{r_2}$, then $a \cup b \in N^{r_1+r_2}$, which holds because $\varphi(a \cup b) = \varphi(a) \cup \varphi(b) \in p^{r_1}(\dots) \cup p^{r_2}(\dots) = p^{r_1+r_2}(\dots)$);
- $q^*(\alpha_r) \in N^{s-r}$ iff $\alpha_r \in N^{s-r}$ (since q^* commutes with φ and hence preserves the Nygaard filtration);
- therefore $q^*(\alpha_r) \cup \xi^r \in N^{s-r} \cup N^r \subseteq N^s$.

Conversely, the Hodge–Tate comparison (property (5)) identifies $\text{gr}_N^s R\Gamma_{\text{Prism}}(\mathbb{P})$ with $H^*(\mathbb{P}, \Omega^s)$ and similarly for the source, and the classical Illusie formula gives an isomorphism at each graded piece, confirming the filtration compatibility. $\square$

3.2. Gysin isomorphism.

Lemma 3.2 (Gysin isomorphism). *Let $i: Z \hookrightarrow X$ be a regular closed immersion of codimension c . There is a canonical isomorphism in $D_{\text{pris}}^{\varphi, N}(A_{\text{inf}})$:*

$$(3) \quad R\Gamma_{\text{Prism}, Z}(X) \simeq R\Gamma_{\text{Prism}}(Z)(-c)[-2c],$$

φ -equivariant and $N^\bullet$ -compatible.

Proof. We give the local computation in detail and then argue globally.

(a) Local model.

Locally on X , the ideal of Z is generated by a regular sequence $f_1, \dots, f_c \in \mathcal{O}_X$. We work in the affine prismatic site of a small affine $U = \text{Spec } A \subset X$ with $Z \cap U = \text{Spec } (A/(f_1, \dots, f_c))$.

The prismatic cohomology with support $R\Gamma_{\text{Prism}, Z \cap U}(U)$ is represented by the complex

$$K^\bullet := \text{Cone}(R\Gamma(U_{\text{Prism}}, \mathcal{O}_{\text{Prism}}) \rightarrow R\Gamma((U \setminus Z \cap U)_{\text{Prism}}, \mathcal{O}_{\text{Prism}}))[-1].$$

Since $f_1, \dots, f_c$ form a regular sequence in A , and the structural sheaf $\mathcal{O}_{\text{Prism}}$ of the prismatic site is flat over A (see [2, Lemma 2.25]), the sequence $f_1, \dots, f_c$ is also regular in $\mathcal{O}_{\text{Prism}}$. The local calculation of cohomology with support via the Koszul complex then gives:

$$R\Gamma_{\text{Prism}, Z \cap U}(U) \simeq \mathcal{O}_{\text{Prism}}/(f_1, \dots, f_c) \cdot \{-c\},$$

where $\{-c\}$ indicates the cohomological shift coming from the Koszul resolution of a length- c regular sequence.

The identification $\mathcal{O}_{\text{Prism}}/(f_1, \dots, f_c) \simeq (i_{Z \cap U})_* \mathcal{O}_{(Z \cap U)_{\text{Prism}}}$ (the structural sheaf of the prismatic site of $Z \cap U$, viewed as a sheaf on U_{Prism}) gives:

$$R\Gamma_{\text{Prism}, Z \cap U}(U) \simeq R\Gamma_{\text{Prism}}(Z \cap U)[-2c]$$

before accounting for the Tate twist.

(b) The Tate twist $(-c)$.

The Tate twist arises from the identification of the top exterior power of the conormal sheaf of Z in X : the prismatic fundamental class $\eta_Z \in H_{\text{Prism}, Z}^{2c}(X)(c)$ (constructed in [2, Section 9] as the class of the closed immersion) corresponds, after the local Koszul computation, to the element $df_1 \wedge \dots \wedge df_c \in (\wedge^c N_{Z/X}^\vee)(c)$. This shifts the Frobenius weight by c , giving the Tate twist $(-c)$ in (3).

Concretely: the Frobenius acts on $\mathcal{O}_{\text{Prism}}/(f_1, \dots, f_c)$ as φ_Z (the Frobenius of Z), while the Tate twist $(-c)$ means that the identification $\text{Koszul}(f_1, \dots, f_c) \simeq (Z \cap U)_{\text{Prism}}$ contributes a factor of p^{-c} to the Frobenius (because $\varphi(f_i) \in f_i^p + p \cdot (\dots)$, and the Koszul computation picks up p^c from the c generators), giving the formula: $\varphi(\eta_Z) = p^c \cdot \eta_Z$, consistent with $\eta_Z \in H_{\text{Prism}, Z}^{2c}(X)(c)$.

(c) Global isomorphism.

The local isomorphisms are compatible on overlaps: on $U \cap V$ for two affines U, V , the two local Koszul computations agree because both are induced by the global excision triangle of Section 2 restricted to $U \cap V$. By the sheaf axiom for the prismatic site, the local isomorphisms glue to a global isomorphism (3).

(d) Nygaard compatibility.

Under the isomorphism (3), the Nygaard filtration on the left corresponds to the filtration on $R\Gamma_{\text{Prism}}(Z)(-c)[-2c]$ as follows. A section $\eta \in R\Gamma_{\text{Prism}, Z}(X)$ lies in N^s iff $\varphi(\eta) \in p^s R\Gamma_{\text{Prism}, Z}(X)$. Under the isomorphism, η corresponds to a class $\eta' \in R\Gamma_{\text{Prism}}(Z)$, and $\varphi(\eta)$ corresponds to $p^c \varphi(\eta')$ (from the Tate twist). So $\eta \in N^s$ iff $p^c \varphi(\eta') \in p^s(\dots)$ iff $\varphi(\eta') \in p^{s-c}(\dots)$ iff $\eta' \in N^{s-c}$. This is precisely the condition that $\eta \in N^s$ corresponds to $\eta' \in N^{s-c}$, i.e., the isomorphism is compatible with $N^s R\Gamma_{\text{Prism}, Z}(X) \simeq N^{s-c} R\Gamma_{\text{Prism}}(Z)$,

which is the Nygaard filtration of $R\Gamma_{\text{Prism}}(Z)(-c)$ (the Tate twist shifts the filtration by $-c$, as in the proof of Lemma 3.1). $\square$

4. PROOF OF THE MAIN THEOREM

Proof of Theorem 1.1. We prove the rational statement first, then address Nygaard compatibility and integrality separately.

Step 1: π^* is split injective.

We claim the composed map $R\Gamma_{\text{Prism}}(X) \xrightarrow{\pi^*} R\Gamma_{\text{Prism}}(\tilde{X}) \xrightarrow{R\pi_*} R\Gamma_{\text{Prism}}(X)$ is the identity.

By the proper base change theorem [1, Theorem 5.2], applied to $\pi: \tilde{X} \rightarrow X$:

$$R\pi_*(R\Gamma_{\text{Prism}}(\tilde{X})) \simeq R\Gamma_{\text{Prism}}(X) \otimes_{\mathcal{O}_X}^L R\pi_*\mathcal{O}_{\tilde{X}}.$$

We need $R\pi_*\mathcal{O}_{\tilde{X}} \simeq \mathcal{O}_X$.

Classical vanishing. Classically (Zariski topology): $\pi_*\mathcal{O}_{\tilde{X}} = \mathcal{O}_X$ because π is birational and X is normal. For $i > 0$: since $Z \hookrightarrow X$ is a regular closed immersion in a regular scheme, and $\tilde{X} = \text{Bl}_Z X$, the classical computation (see [5, Exposé VII, Prop. 3.3]) gives $R^i\pi_*\mathcal{O}_{\tilde{X}} = 0$ for $i > 0$. More precisely: locally, π is the blow-up of $\text{Spec } A$ along a regular ideal $(f_1, \dots, f_c)$, and the Leray spectral sequence for π together with the explicit description of $\mathcal{O}_{\tilde{X}}$ via the Rees algebra $\bigoplus_{n \geq 0} I^n$ (where $I = (f_1, \dots, f_c)$) gives $H^i(\text{Bl}_I(\text{Spec } A), \mathcal{O}) = 0$ for $0 < i < c - 1$ by a direct Koszul computation; for a regular center ($c \geq 2$) the vanishing extends to all $i > 0$. Therefore $R\pi_*\mathcal{O}_{\tilde{X}} = \mathcal{O}_X$ as complexes of $\mathcal{O}_X$ -modules.

Transfer to the prismatic setting via two comparisons. We spell out the chain of isomorphisms explicitly. Let $\alpha_c: R\Gamma_{\text{Prism}}(-) \otimes_{A_{\text{inf}}} A_{\text{cris}} \xrightarrow{\sim} R\Gamma_{\text{cris}}(-/W(k))$ be the crystalline comparison of Section 2(3).

(i) Applying $(-) \otimes_{A_{\text{inf}}} A_{\text{cris}}$ to both sides of the equation $R\pi_* R\Gamma_{\text{Prism}}(\tilde{X}) \otimes_{A_{\text{inf}}} A_{\text{cris}}$, and using that α_c commutes with proper pushforward (both α_c and $R\pi_*$ commute because they are natural transformations of cohomological functors on the same underlying scheme; see [1, Corollary 5.3] for the prismatic pushforward formula):

$$R\pi_* R\Gamma_{\text{Prism}}(\tilde{X}) \otimes_{A_{\text{inf}}} A_{\text{cris}} \simeq R\pi_* R\Gamma_{\text{cris}}(\tilde{X}/W(k)).$$

(ii) In crystalline cohomology, proper pushforward is compatible with the comparison [3, Theorem 1.4], and the classical vanishing $R^i\pi_*\mathcal{O}_{\tilde{X}} = 0$ implies (via the Berthelot–Ogus comparison for the blow-up):

$$R\pi_* R\Gamma_{\text{cris}}(\tilde{X}/W(k)) \simeq R\Gamma_{\text{cris}}(X/W(k)).$$

(iii) Applying α_c^{-1} in the reverse direction:

$$R\pi_* R\Gamma_{\text{Prism}}(\tilde{X}) \otimes_{A_{\text{inf}}} A_{\text{cris}} \simeq R\Gamma_{\text{Prism}}(X) \otimes_{A_{\text{inf}}} A_{\text{cris}}.$$

(iv) By faithful flatness of $A_{\text{inf}}[1/p] \rightarrow A_{\text{cris}}[1/p]$ [1, Lemma 3.23], an isomorphism after tensoring with $A_{\text{cris}}[1/p]$ implies an isomorphism after tensoring with $A_{\text{inf}}[1/p]$, giving:

$$R\pi_* R\Gamma_{\text{Prism}}(\tilde{X})[1/p] \simeq R\Gamma_{\text{Prism}}(X)[1/p].$$

In particular, $(R\pi_*) \circ (\pi^*) = \text{id}$ in $D_{\text{pris}}^{\varphi, N}(A_{\text{inf}}[1/p])$, so π^* is split injective with complement:

$$C := \text{Cone}(\pi^*)[-1] \in D_{\text{pris}}^{\varphi, N}(A_{\text{inf}}[1/p]).$$

Step 2: Identification $C \simeq R\Gamma_{\text{Prism}, E}(\tilde{X})$.

The excision triangle for $(\tilde{X}, E)$ (Remark 2.1) is:

$$R\Gamma_{\text{Prism}, E}(\tilde{X}) \rightarrow R\Gamma_{\text{Prism}}(\tilde{X}) \rightarrow R\Gamma_{\text{Prism}}(\tilde{X} \setminus E) \xrightarrow{+1}.$$

Since $\pi|_{\tilde{X} \setminus E}$ is an isomorphism onto $X \setminus Z$, we have $R\Gamma_{\text{Prism}}(\tilde{X} \setminus E) \simeq R\Gamma_{\text{Prism}}(X \setminus Z)$. Comparing with the excision triangle for (X, Z) :

$$R\Gamma_{\text{Prism}, Z}(X) \rightarrow R\Gamma_{\text{Prism}}(X) \xrightarrow{j^*} R\Gamma_{\text{Prism}}(X \setminus Z) \xrightarrow{+1},$$

and using the splitting $R\Gamma_{\text{Prism}}(\tilde{X}) = R\Gamma_{\text{Prism}}(X) \oplus C$ from Step 1, we get a commutative diagram in $D_{\text{pris}}^{\varphi, N}(A_{\text{inf}}[1/p])$:

$$\begin{array}{ccccc} R\Gamma_{\text{Prism}, E}(\tilde{X}) & \rightarrow & R\Gamma_{\text{Prism}}(X) \oplus C & \rightarrow & R\Gamma_{\text{Prism}}(X \setminus Z) \\ \downarrow \wr & & \downarrow & & \downarrow \wr \\ C & \rightarrow & R\Gamma_{\text{Prism}}(X) \oplus C & \rightarrow & R\Gamma_{\text{Prism}}(X \setminus Z) \end{array}$$

The first vertical map is an isomorphism by the five lemma in the triangulated category $D_{\text{pris}}^{\varphi, N}(A_{\text{inf}}[1/p])$: both rows are distinguished triangles, the rightmost map is an isomorphism, and the middle map is the identity. Hence $C \simeq R\Gamma_{\text{Prism}, E}(\tilde{X})$.

Step 3: Computation of C .

Apply Lemma 3.2 to the divisor $E \hookrightarrow \tilde{X}$ (codimension 1):

$$R\Gamma_{\text{Prism}, E}(\tilde{X}) \simeq R\Gamma_{\text{Prism}}(E)(-1)[-2].$$

Apply Lemma 3.1 to the bundle $q: E = \mathbb{P}(N_{Z/X}) \rightarrow Z$ of rank c :

$$R\Gamma_{\text{Prism}}(E) \simeq \bigoplus_{r=0}^{c-1} R\Gamma_{\text{Prism}}(Z)(-r)[-2r].$$

Combining and re-indexing:

$$(4) \quad C \simeq R\Gamma_{\text{Prism}}(E)(-1)[-2] \simeq \bigoplus_{r=0}^{c-1} R\Gamma_{\text{Prism}}(Z)(-r-1)[-2r-2] = \bigoplus_{r=1}^c R\Gamma_{\text{Prism}}(Z)(-r)[-2r].$$

Step 4: Elimination of the $r = c$ summand.

We claim the summand $r = c$ in (4), namely $R\Gamma_{\text{Prism}}(Z)(-c)[-2c]$, lies in the image of π^* and is not part of C .

By Lemma 3.2 applied to $i: Z \hookrightarrow X$: $R\Gamma_{\text{Prism}, Z}(X) \simeq R\Gamma_{\text{Prism}}(Z)(-c)[-2c]$. The natural map $R\Gamma_{\text{Prism}, Z}(X) \rightarrow R\Gamma_{\text{Prism}}(X)$ (the first arrow of the excision triangle for (X, Z)) factors through π^* : schematically, the class of Z in X pulls back to the class of E in $\tilde{X}$ via π , and E maps into $R\Gamma_{\text{Prism}, E}(\tilde{X})$. The $r = 0$ piece of the projective bundle decomposition of E (the piece $q^*(\alpha) \cup \xi^0 = q^*(\alpha)$, i.e., the pullback from Z) pushes forward via π_* back to $R\Gamma_{\text{Prism}}(X)$:

$$R\pi_*(q^*(\alpha)) = \alpha \cdot R\pi_*\mathcal{O}_E = \alpha \cdot i_*\mathcal{O}_Z \in R\Gamma_{\text{Prism}}(X),$$

where the projection formula and $R\pi_*\mathcal{O}_E = i_*\mathcal{O}_Z$ (the restriction of $\pi_*\mathcal{O}_{\tilde{X}} = \mathcal{O}_X$ to Z) are used. Hence this piece does not contribute to C , and we conclude:

$$C \simeq \bigoplus_{r=1}^{c-1} R\Gamma_{\text{Prism}}(Z)(-r)[-2r],$$

completing the rational decomposition (1). □

5. NYGAARD COMPATIBILITY

Proposition 5.1 (Nygaard compatibility of the blow-up formula). *The isomorphism (1) of Theorem 1.1 is compatible with the Nygaard filtration. Precisely, for each $s \geq 0$, the induced map on N^s -filtration pieces is an isomorphism:*

$$(5) \quad N^s R\Gamma_{\text{Prism}}(\tilde{X}) \simeq N^s R\Gamma_{\text{Prism}}(X) \oplus \bigoplus_{r=1}^{c-1} N^{s+r} R\Gamma_{\text{Prism}}(Z).$$

The Tate twist convention is: $N^s(M(-r)) = N^{s+r}(M)$, i.e. a negative twist increases the Nygaard filtration level required. Equivalently, $N^{s+r} R\Gamma_{\text{Prism}}(Z)$ is the subspace of sections whose Frobenius satisfies $\varphi(\alpha) \in p^{s+r} R\Gamma_{\text{Prism}}(Z)$, and such a section lies in N^s of $R\Gamma_{\text{Prism}}(Z)(-r)$ because the $(-r)$ twist scales φ down by p^r .

Proof. We verify (5) at each step of the proof of Theorem 1.1, using the Tate twist convention $N^s(M(-r)) = N^{s+r}(M)$ throughout.

Convention check. If $\varphi_{M(-r)} = p^{-r} \varphi_M$, then:

$$x \in N^s(M(-r)) \iff p^{-r} \varphi_M(x) \in p^s M \iff \varphi_M(x) \in p^{s+r} M \iff x \in N^{s+r}(M).$$

So indeed $N^s(M(-r)) = N^{s+r}(M)$: the $(-r)$ twist raises the required Nygaard level by r . Equivalently, the Nygaard filtration of $M(-r)$ is indexed so that what was N^{s+r} of M becomes N^s of $M(-r)$.

Step 1 (splitting of π^*). The maps π^* and $R\pi_*$ preserve $N^\bullet$: π^* preserves N^s (as it commutes with φ); $R\pi_*$ preserves it by [1, Theorem 5.2]. The splitting $R\Gamma_{\text{Prism}}(\tilde{X}) \simeq R\Gamma_{\text{Prism}}(X) \oplus C$ is therefore $N^\bullet$ -compatible, giving $N^s R\Gamma_{\text{Prism}}(\tilde{X}) \simeq N^s R\Gamma_{\text{Prism}}(X) \oplus N^s C$.

Step 2 (identification of C). The two excision triangles (for $(\tilde{X}, E)$ and for (X, Z)) are both $N^\bullet$ -compatible by Remark 2.1. The diagram of filtered triangles:

$$\begin{array}{ccccc} R\Gamma_{\text{Prism}, E}(\tilde{X}) & \rightarrow & R\Gamma_{\text{Prism}}(X) \oplus C & \rightarrow & R\Gamma_{\text{Prism}}(X \setminus Z) \\ \downarrow f & & \downarrow \text{id} & & \downarrow \text{id} \\ C & \rightarrow & R\Gamma_{\text{Prism}}(X) \oplus C & \rightarrow & R\Gamma_{\text{Prism}}(X \setminus Z) \end{array}$$

has both rows as distinguished filtered triangles, with the two right vertical maps being isomorphisms. By the five lemma in filtered triangulated categories (for bounded filtrations, the filtered five lemma follows from comparing graded pieces, where the five lemma in the abelian category of $W(k)$ -modules applies), f is an isomorphism of filtered complexes.

Step 3 (Gysin and projective bundle). By Lemma 3.2(d), applying the Tate twist convention with $r = 1$:

$$N^s R\Gamma_{\text{Prism}, E}(\tilde{X}) \simeq N^s(R\Gamma_{\text{Prism}}(E)(-1)) = N^{s+1} R\Gamma_{\text{Prism}}(E).$$

By Lemma 3.1(d), the projective bundle isomorphism gives (applying N^{s+1} to $R\Gamma_{\text{Prism}}(E)$, which itself decomposes as $\bigoplus_{r=0}^{c-1} R\Gamma_{\text{Prism}}(Z)(-r)$):

$$N^{s+1} R\Gamma_{\text{Prism}}(E) = N^{s+1} \left(\bigoplus_{r=0}^{c-1} R\Gamma_{\text{Prism}}(Z)(-r) \right) = \bigoplus_{r=0}^{c-1} N^{s+1}(R\Gamma_{\text{Prism}}(Z)(-r)) = \bigoplus_{r=0}^{c-1} N^{s+1+r} R\Gamma_{\text{Prism}}(Z)$$

Re-indexing with $r' = r + 1$ (so $r' = 1, \dots, c$):

$$N^s R\Gamma_{\text{Prism}, E}(\tilde{X}) \simeq \bigoplus_{r'=1}^c N^{s+r'} R\Gamma_{\text{Prism}}(Z).$$

Step 4 ($r = c$ elimination). The $r' = c$ summand is $N^{s+c}R\Gamma_{\text{Prism}}(Z)$. By Lemma 3.2(d) applied to the Gysin isomorphism $R\Gamma_{\text{Prism},Z}(X) \simeq R\Gamma_{\text{Prism}}(Z)(-c)[-2c]$:

$$N^s R\Gamma_{\text{Prism},Z}(X) = N^s(R\Gamma_{\text{Prism}}(Z)(-c)) = N^{s+c}R\Gamma_{\text{Prism}}(Z).$$

Hence the $r' = c$ summand is exactly $N^s R\Gamma_{\text{Prism},Z}(X)$, which maps into $N^s R\Gamma_{\text{Prism}}(X)$ via the filtered excision map. It therefore contributes to $N^s R\Gamma_{\text{Prism}}(X)$, not to $N^s C$.

The remaining summands $r' = 1, \dots, c-1$ give:

$$N^s C \simeq \bigoplus_{r=1}^{c-1} N^{s+r} R\Gamma_{\text{Prism}}(Z),$$

which is precisely (5). $\square$

6. INTEGRAL DESCENT AND TORSION BOUNDS

Proposition 6.1 (Criterion: rational $\Rightarrow$ integral). *Suppose $X/\mathcal{O}_K$ satisfies:*

- (A) *X admits a smooth lifting over $W_2(k)$ with a locally compatible Frobenius lift;*
- (B) *all $H^j(X, \Omega_{X/\mathcal{O}_K}^j)$ are p -torsion-free.*

Then any isomorphism in $D_{\text{pris}}^{\varphi,N}(A_{\text{inf}}[1/p])$ between perfect prismatic complexes is already an isomorphism in $D_{\text{pris}}^{\varphi,N}(A_{\text{inf}})$.

Proof. Let $f: P \rightarrow Q$ be a map of perfect A_{inf} -complexes that is an isomorphism after inverting p . Set $C_f = \text{Cone}(f)$. We must show $H^i(C_f) = 0$ for all i .

Step 1: Reduction to mod p .

Since $C_f[1/p] = 0$, each $H^i(C_f)$ is a finitely generated A_{inf} -module killed by p^N for some N . We claim it suffices to show $H^i(C_f)/p = 0$.

Indeed: $H^i(C_f)/p = 0$ means $H^i(C_f) = p \cdot H^i(C_f)$. Iterating: $H^i(C_f) = p \cdot H^i(C_f) = p^2 \cdot H^i(C_f) = \dots = p^N \cdot H^i(C_f) = 0$, where the last equality uses $p^N H^i(C_f) = 0$. (This is a basic consequence of the p -adic filtration and does not require Nakayama's lemma.)

Step 2: Vanishing mod p via graded pieces.

We compute $H^i(C_f)/p = H^i(C_f \otimes_{A_{\text{inf}}}^L A_{\text{inf}}/p)$.

The Nygaard filtration on P and Q has graded pieces

$$\text{gr}_N^j H^i(P) \otimes_{W(k)} k \simeq H^{i-j}(X, \Omega_{X/\mathcal{O}_K}^j) \otimes_{W(k)} k,$$

which is a k -vector space of finite dimension (property (5) of Section 2). By hypothesis (B), $H^{i-j}(X, \Omega_{X/\mathcal{O}_K}^j)$ is p -torsion-free, hence free over $W(k)$ (finitely generated torsion-free modules over the DVR $W(k)$ are free). Similarly for the graded pieces of Q .

The map $\text{gr}_N^j f$ is therefore a map between free $W(k)$ -modules of finite rank. Since $f[1/p]$ is an isomorphism, $\text{gr}_N^j f \otimes \text{Frac}(W(k))$ is an isomorphism. A map between free $W(k)$ -modules of the same finite rank that becomes an isomorphism after inverting p (i.e., after tensoring with $W(k)[1/p]$) is already an isomorphism over $W(k)$: its determinant is a unit in $W(k)$ (because the determinant is rational and also lies in $W(k)$, hence is a p -adic unit). Thus $\text{gr}_N^j f$ is an isomorphism over $W(k)$, and in particular $\text{gr}_N^j H^i(C_f) \otimes k = 0$.

Step 3: Nygaard degeneration gives $H^i(C_f)/p = 0$.

By hypothesis (A) and the theorem of Deligne–Illusie [4], the Hodge spectral sequence degenerates at E_1 integrally over $W_2(k)$ (and hence over $W(k)$ by induction, via the deformation theory of [4, Remark 2.5]). By property (5) of Section 2, this implies the Nygaard spectral sequence

$$E_1^{j,i-j} = \text{gr}_N^j H^i(-) \otimes k \Rightarrow H^i(-) \otimes k$$

degenerates at E_1 for P and for Q . Hence:

$$H^i(P) \otimes k = \bigoplus_j \mathrm{gr}_N^j H^i(P) \otimes k, \quad H^i(Q) \otimes k = \bigoplus_j \mathrm{gr}_N^j H^i(Q) \otimes k.$$

Since $\mathrm{gr}_N^j f$ is an isomorphism (Step 2), the map $H^i(f) \otimes k: H^i(P) \otimes k \rightarrow H^i(Q) \otimes k$ is an isomorphism on each graded piece, hence an isomorphism. Therefore $H^i(C_f) \otimes k = H^i(C_f \otimes^L k) = 0$, i.e., $H^i(C_f)/p = 0$.

By Step 1, $H^i(C_f) = 0$ for all i .

Conclusion. Both X and Z satisfy (A)+(B) (hypothesis (A) for Z follows because the Frobenius lift on X restricts to one on Z , and (B) for Z is given). Hence the argument above applies to each summand of (1), and the rational isomorphism is integral. $\square$

Proposition 6.2 (Torsion bounds). *Without assuming (A)+(B), suppose that for all j , the groups $H^{i-j}(X, \Omega^j)$ and $H^{i-2r-j}(Z, \Omega^j)$ are annihilated by p^{t_j} . Then $H_{\mathrm{Prism}}^i(\tilde{X})[p^\infty]$ is annihilated by p^N with*

$$N \leq \max(N_X, \max_{1 \leq r \leq c-1} N_{Z,i-2r}),$$

where $N_X = \sum_j t_j^X + C_X$ and $N_{Z,m} = \sum_j t_j^Z + C_Z$ with C_X, C_Z depending only on $\dim X$, $\dim Z$, and the Hodge numbers, via the spectral sequence bound of [1, Prop. 4.13]. Under (A)+(B), $N = 0$.

Proof. Immediate from the direct sum decomposition (1) together with the torsion estimates for each summand from [1, Prop. 4.13]. Under (A)+(B), Proposition 6.1 gives $N = 0$. $\square$

7. EXAMPLES

Example 7.1 ($\mathrm{Bl}_x \mathbb{P}^2$). Let $X = \mathbb{P}_{\mathcal{O}_K}^2$, $Z = \{x\}$ a rational $\mathcal{O}_K$ -point, $c = 2$. Then $R\Gamma_{\mathrm{Prism}}(Z) \simeq A_{\mathrm{inf}}$ (a point), and (1) gives:

$$R\Gamma_{\mathrm{Prism}}(\mathrm{Bl}_x \mathbb{P}^2) \simeq R\Gamma_{\mathrm{Prism}}(\mathbb{P}^2) \oplus A_{\mathrm{inf}}(-1)[-2].$$

In degree 2: $H_{\mathrm{Prism}}^2(\mathrm{Bl}_x \mathbb{P}^2) \simeq H_{\mathrm{Prism}}^2(\mathbb{P}^2) \oplus A_{\mathrm{inf}}(-1)$.

Verification of the Nygaard filtration (Proposition 5.1):

- $N^1 H_{\mathrm{Prism}}^2(\mathbb{P}^2) = H_{\mathrm{Prism}}^2(\mathbb{P}^2)$ (since $H^2(\mathbb{P}^2, \mathcal{O}) = 0$ forces $\mathrm{gr}_N^0 H^2 = 0$ and everything lies in N^1);
- the summand $A_{\mathrm{inf}}(-1)$ satisfies $N^1(A_{\mathrm{inf}}(-1)) = A_{\mathrm{inf}}(-1)$ and $N^2(A_{\mathrm{inf}}(-1)) = 0$ (from $\mathrm{gr}_N^1 = A_{\mathrm{inf}}$, consistent with $c_1^{\mathrm{Prism}}(E) \in N^1$);
- φ acts on $A_{\mathrm{inf}}(-1)$ by p , consistent with $\varphi(c_1^{\mathrm{Prism}}(E)) = p \cdot c_1^{\mathrm{Prism}}(E)$.

Both $X = \mathbb{P}^2$ and $Z = \{x\}$ satisfy (A)+(B), so the formula is integral with $N = 0$.

Example 7.2 (Kummer K3). Let $A = E_1 \times E_2$ be a product of elliptic curves over $\mathcal{O}_K$ with good reduction ($p > 2$). The Kummer surface $K = \mathrm{Kum}(A)$ is obtained by blowing up the 16 singular points of $A/(\pm 1)$. Applying (1) at each of the 16 centers (all $c = 2$):

$$R\Gamma_{\mathrm{Prism}}(K) \simeq R\Gamma_{\mathrm{Prism}}(A/\mu)^{\mathrm{sm}} \oplus \bigoplus_{i=1}^{16} A_{\mathrm{inf}}(-1)[-2].$$

The Nygaard filtration: each $A_{\mathrm{inf}}(-1)[-2]$ summand lies in $N^1 H_{\mathrm{Prism}}^2(K)$ (from the Tate twist), consistent with the exceptional classes being algebraic of Hodge type $(1, 1)$. Under (A)+(B) (which hold for K3 surfaces with $p > 2$ by the deformation theory of [4]), the formula is integral and the p -torsion exponent is 0.

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THE PRISMATIC CYCLE MAP: CONSTRUCTION AND LOCAL-TO-GLOBAL SURJECTIVITY *REVISED VERSION*

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ABSTRACT. This is the second paper in a series devoted to global prismatic geometry. Building on the blow-up formula established in Part I [II], we construct the prismatic cycle map

$$\mathrm{cl}_{\mathrm{Prism}}^r: \mathcal{Z}^r(X) \longrightarrow H_{\mathrm{Prism}}^{2r}(X)(r),$$

which associates to each algebraic cycle of codimension r a class in prismatic cohomology with Tate twist. We prove its basic properties: functoriality, compatibility with the crystalline and de Rham cycle maps via the comparison isomorphisms, and φ -equivariance.

Our main result is a local-to-global surjectivity theorem: if a prismatic cohomology class $\alpha \in H_{\mathrm{Prism}}^{2r}(X)(r)$ is locally (on a finite open cover) in the image of $\mathrm{cl}_{\mathrm{Prism}}^r$, then α is globally in the image of $\mathrm{cl}_{\mathrm{Prism}}^r$, possibly after multiplication by a bounded p -power whose exponent is controlled by the torsion estimates of Part I. For $r = 1$ (the divisor case) the statement holds without any p -power correction, giving a prismatic analogue of classical Picard gluing.

Key revisions in this version. Three gaps identified in the previous version have been addressed: (1) Remark 2.2 now supplies a self-contained proof that the fundamental class is independent of the choice of resolution, via correspondences and the de Jong theorem; (2) Lemma 3.1 establishes that $\mathcal{F}^r$ is a well-defined sheaf of abelian groups and that the vanishing of $\omega(\{Z_i\})$ characterises global gluing; (3) the proof of Theorem 4.1 gives explicit, mutually exhaustive conditions for Strategies A, B, and C, and Strategy C is completed with a direct cycle-level argument showing that cohomological vanishing implies algebraic gluing.

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1. INTRODUCTION

A cohomological cycle map is a fundamental bridge between algebra and geometry: it sends an algebraic cycle—a formal combination of irreducible subvarieties—to a cohomological class encoding its arithmetic and geometric properties. In singular cohomology the cycle map is the foundation of Hodge theory; in étale cohomology it underlies the Tate conjecture; in de Rham cohomology it appears in the work of Grothendieck and Deligne.

In this paper we construct the corresponding map for prismatic cohomology and establish its first global property. Since prismatic cohomology interpolates between the p -adic realizations (crystalline, de Rham, étale) in a single integral object, a prismatic cycle map provides a unified framework for studying cycle classes across all these theories simultaneously.

Setup and notation. Let p be a prime, K a complete p -adic field with ring of integers $\mathcal{O}_K$ and perfect residue field k . Let X be a regular scheme, proper and smooth over $\mathcal{O}_K$. We write $H_{\text{Prism}}^i(X) = H^i(X_{\text{Prism}}, \mathcal{O}_{\text{Prism}})$ for the prismatic cohomology groups, φ for the Frobenius, and $N^\bullet$ for the Nygaard filtration. We refer to Part I [1] for the basic formalism and notation; in particular, the derived category $D_{\text{pris}}^{\varphi, N}(A_{\text{inf}})$ and the comparison isomorphisms are as in [1] (Section 2 therein).

We denote by $\mathcal{Z}^r(X)$ the group of algebraic cycles of codimension r on X (finite $\mathbb{Z}$ -linear combinations of integral closed subschemes of codimension r).

Main results.

Theorem 1.1 (Prismatic cycle map; see Theorem 2.3). *Let X be regular, proper, smooth over $\mathcal{O}_K$, and let $r \geq 1$. There exists a canonical $\mathbb{Z}$ -linear map (the prismatic cycle map)*

$$\text{cl}_{\text{Prism}}^r : \mathcal{Z}^r(X) \longrightarrow H_{\text{Prism}}^{2r}(X)(r),$$

with the following properties:

- (1) (Functoriality) $\text{cl}_{\text{Prism}}^r$ is compatible with proper pushforward and flat pullback of cycles.
- (2) (Comparison) Under the isomorphisms $H_{\text{Prism}}^{2r}(X)(r) \otimes_{A_{\text{inf}}} A_{\text{cris}} \simeq H_{\text{cris}}^{2r}(X)(r)$ and $H_{\text{Prism}}^{2r}(X)(r) \otimes_{A_{\text{inf}}} B_{\text{dR}}^+ \simeq H_{\text{dR}}^{2r}(X)$, the prismatic cycle map recovers the crystalline and de Rham cycle maps.
- (3) (φ -equivariance) $\varphi(\text{cl}_{\text{Prism}}^r([Z])) = p^r \text{cl}_{\text{Prism}}^r([Z])$ for any cycle $[Z] \in \mathcal{Z}^r(X)$.
- (4) (Nygaard degree) $\text{cl}_{\text{Prism}}^r([Z]) \in N^r H_{\text{Prism}}^{2r}(X)(r)$, i.e., cycle classes lie in the correct Nygaard filtration degree.

Theorem 1.2 (Local-to-global surjectivity; see Theorem 4.1). *Let X be as above and $\alpha \in H_{\text{Prism}}^{2r}(X)(r)$. Suppose there exists a finite open cover $\{U_i\}_{i \in I}$ of X such that for each i , the restriction $\alpha|_{U_i}$ belongs to the image of $\text{cl}_{\text{Prism}}^r : \mathcal{Z}^r(U_i) \rightarrow H_{\text{Prism}}^{2r}(U_i)(r)$. Then:*

- (1) For $r = 1$: there exists a global cycle $Z \in \mathcal{Z}^r(X)$ with $\text{cl}_{\text{Prism}}^1([Z]) = \alpha$. (No p -power correction needed.)
- (2) For $r \geq 2$: there exists $N \geq 0$ and a global cycle $Z \in \mathcal{Z}^r(X)$ with $\text{cl}_{\text{Prism}}^r([Z]) = p^N \alpha$. The exponent N is bounded explicitly in terms of the p -torsion of $H_{\text{Prism}}^{2r-1}(X)(r)$, which is controlled by Proposition 4.2 of Part I [1].

Relation to the Hodge conjecture. The classical Hodge conjecture asks whether every rational (r, r) -Hodge class on a smooth complex projective variety is algebraic. In the p -adic setting the Tate conjecture makes an analogous assertion for étale cohomology. Theorem 1.2 establishes a local-to-global version of these conjectures in prismatic cohomology: local algebraicity implies global algebraicity up to a controlled p -power. While this falls short of proving the Hodge or Tate conjectures, it provides the structural result needed for the regulator construction in Part III of this series.

What is new in this paper.

- (1) The construction of $\mathrm{cl}_{\mathrm{Prism}}^r$ via Gysin maps and prismatic Chern classes of 3 is natural and expected; to our knowledge it has not been given a complete self-contained treatment.
- (2) The local-to-global surjectivity theorem (Theorem 1.2) and its explicit p -power bound appear to be new.
- (3) The use of the blow-up formula of Part I 1 to track cycle classes under resolution of singularities is a new application of that result.

Organization. Section 2 constructs the prismatic cycle map and proves Theorem 1.1. Section 3 develops the Čech obstruction theory for the gluing problem. Section 4 proves Theorem 1.2. Section 5 works through examples. Section 6 describes the connection to Part III (prismatic regulators).

2. CONSTRUCTION OF THE PRISMATIC CYCLE MAP

2.1. The fundamental class of a regular subscheme. Let $i: Z \hookrightarrow X$ be a regular closed immersion of codimension r . Recall from Part I, Lemma 3.2, the Gysin isomorphism:

$$\mathrm{R}\Gamma_{\mathrm{Prism}, Z}(X) \simeq \mathrm{R}\Gamma_{\mathrm{Prism}}(Z)(-r)[-2r].$$

Taking cohomology in degree 0 on the right-hand side yields $H_{\mathrm{Prism}}^0(Z) \otimes A_{\mathrm{inf}}(-r)$. The unit element $1 \in H_{\mathrm{Prism}}^0(Z) = A_{\mathrm{inf}}$ (the fundamental class of Z over itself) maps under the Gysin isomorphism and the natural map $H_{\mathrm{Prism}, Z}^{2r}(X)(r) \rightarrow H_{\mathrm{Prism}}^{2r}(X)(r)$ to a class

$$\eta_Z \in H_{\mathrm{Prism}}^{2r}(X)(r),$$

which we call the *prismatic fundamental class* of Z in X .

Definition 2.1 (Prismatic cycle map). *The prismatic cycle map*

$$\mathrm{cl}_{\mathrm{Prism}}^r: \mathcal{Z}^r(X) \longrightarrow H_{\mathrm{Prism}}^{2r}(X)(r)$$

is defined by $\mathbb{Z}$ -linear extension of the assignment $[Z] \mapsto \eta_Z$, where $Z \subset X$ ranges over integral closed subschemes of codimension r . When Z is not regularly embedded in X , we first choose an alteration $\tilde{Z} \rightarrow Z$ with $\tilde{Z}$ regular (which exists by de Jong's theorem 5), and set $\eta_Z := (\tilde{Z} \hookrightarrow X)_[1_{\tilde{Z}}]$ via the proper pushforward of the fundamental class; the result is independent of the choice of $\tilde{Z}$ by Remark 2.2.*

Remark 2.2 (Independence of the resolution). *We give a direct proof that η_Z is independent of the choice of alteration $\tilde{Z} \rightarrow Z$.*

Step 1: Two resolutions are dominated by a third. Let $f_1: \tilde{Z}_1 \rightarrow Z$ and $f_2: \tilde{Z}_2 \rightarrow Z$ be two alterations with $\tilde{Z}_1, \tilde{Z}_2$ regular. By de Jong's theorem 5 applied to the fibre product $\tilde{Z}_1 \times_Z \tilde{Z}_2$, there exists an alteration $g: \tilde{Z}_3 \rightarrow \tilde{Z}_1 \times_Z \tilde{Z}_2$ with $\tilde{Z}_3$ regular, giving a commutative diagram

$$\tilde{Z}_3 \xrightarrow{g_1} \tilde{Z}_1 \xrightarrow{f_1} Z \hookrightarrow X, \quad \tilde{Z}_3 \xrightarrow{g_2} \tilde{Z}_2 \xrightarrow{f_2} Z \hookrightarrow X.$$

It therefore suffices to show that $(f_1 \circ g_1)_*[1_{\tilde{Z}_3}] = (f_1)_*[1_{\tilde{Z}_1}]$ in $H_{\text{Prism}}^{2r}(X)(r)$ (and the symmetric statement for f_2).

Step 2: Behaviour under an alteration between resolutions. Let $g: \tilde{Z}_3 \rightarrow \tilde{Z}_1$ be a proper surjective generically finite morphism of regular schemes of degree d . Because the fundamental class is functorial under proper pushforward (Part I, Section 2.4) and the projection formula gives

$$g_*[1_{\tilde{Z}_3}] = d \cdot [1_{\tilde{Z}_1}] \quad \text{in } H_{\text{Prism}}^0(\tilde{Z}_1),$$

we obtain

$$(f_1 \circ g)_*[1_{\tilde{Z}_3}] = (f_1)_*(g_*[1_{\tilde{Z}_3}]) = d \cdot (f_1)_*[1_{\tilde{Z}_1}] \in H_{\text{Prism}}^{2r}(X)(r).$$

Thus both expressions compute $d \cdot \eta_Z$; they agree after tensoring with $\mathbb{Q}$ (i.e., after inverting p).

Step 3: Integrality via the blow-up formula. To promote the comparison from $\mathbb{Q}$ to $\mathbb{Z}$, we use the fact that f_1 and f_2 can each be factored, up to a further alteration of degree prime to p , as a tower of blow-ups along regularly embedded centres (this is the strong form of de Jong's theorem ensuring alterations of degree a power of p composed with blow-up towers; see [5] Theorem 4.1 and [6]). For each blow-up step $\pi: \text{Bl}_W V \rightarrow V$ along a regularly embedded centre W , the blow-up formula (Part I, Theorem 4.1) gives a canonical splitting $R\Gamma_{\text{Prism}}(\text{Bl}_W V) \simeq R\Gamma_{\text{Prism}}(V) \oplus \bigoplus_{s=1}^{c-1} R\Gamma_{\text{Prism}}(W)(-s)[-2s]$, showing that $\pi_*[1_{\text{Bl}_W V}] = [1_V]$ at the level of fundamental classes (the extra summands carry no degree-0 class). Applying this inductively through the blow-up tower shows that the pushforward of the fundamental class is invariant.

Combining Steps 1–3, the class $(f_1)_*[1_{\tilde{Z}_1}]$ and $(f_2)_*[1_{\tilde{Z}_2}]$ both coincide, integrally, with the common class η_Z . $\square$

2.2. Proof of Theorem 1.1.

Theorem 2.3 (Properties of the prismatic cycle map). *The map $\text{cl}_{\text{Prism}}^r$ of Definition 2.1 satisfies properties (1)–(4) of Theorem 1.1.*

Proof. (1) *Functoriality.* For proper pushforward: if $f: X' \rightarrow X$ is proper and $Z' \subset X'$ is integral of codimension r , then $f_*([Z']) = \deg(Z' \rightarrow f(Z')) \cdot [f(Z')]$ as cycles, and $f_*(\eta_{Z'}) = \eta_{f(Z')}$ in prismatic cohomology by the proper base change theorem [2] and the φ -equivariance of the Gysin map.

For flat pullback: if $f: X' \rightarrow X$ is flat, then $f^*([Z]) = [f^{-1}(Z)]$ as cycles (for Z regular in X), and $f^*(\eta_Z) = \eta_{f^{-1}(Z)}$ by the compatibility of the Gysin map with pullback [3], Section 9.

(2) *Comparison.* Under the comparison $R\Gamma_{\text{Prism}}(X) \otimes_{A_{\text{inf}}} A_{\text{cris}} \simeq R\Gamma_{\text{cris}}(X)$ of [2], the prismatic Gysin map corresponds to the crystalline Gysin map, and $\eta_Z \otimes 1$ coincides with the crystalline fundamental class of Z . Similarly for the de Rham comparison via $\otimes_{A_{\text{inf}}} B_{\text{dR}}^+$.

(3) *φ -equivariance.* The Gysin isomorphism is φ -equivariant (Part I, Lemma 3.2), and the Frobenius acts on the Tate twist (r) by the factor p^r . Therefore $\varphi(\eta_Z) = p^r \eta_Z$ for any cycle Z .

(4) *Nygaard degree.* The class η_Z is the image of the unit in $H_{\text{Prism}}^0(Z) = A_{\text{inf}}$ under the Gysin map. By the Nygaard compatibility of the Gysin isomorphism (Part I, Lemma 3.2), this unit lives in $N^0 A_{\text{inf}} = A_{\text{inf}}$, and its image under the Gysin twist $(-r)[-2r]$ lands in $N^r H_{\text{Prism}}^{2r}(X)(r)$, as required. $\square$

3. THE ČECH OBSTRUCTION TO GLOBAL GLUING

Let $\alpha \in H_{\text{Prism}}^{2r}(X)(r)$ and suppose a finite cover $\{U_i\}$ is given such that $\alpha|_{U_i} = \text{cl}_{\text{Prism}}^r([Z_i])$ for cycles $Z_i \in \mathcal{Z}^r(U_i)$.

3.1. The sheaf $\mathcal{F}^r$.

Lemma 3.1 ($\mathcal{F}^r$ is a sheaf). *The presheaf*

$$\mathcal{F}^r(U) := \ker(\text{cl}_{\text{Prism}}^r : \mathcal{Z}^r(U) \rightarrow H_{\text{Prism}}^{2r}(U)(r))$$

on the Zariski site of X is a sheaf of abelian groups. Moreover, $\text{cl}_{\text{Prism}}^r$ is a morphism of presheaves, i.e., for every open immersion $V \hookrightarrow U$ the square

$$\begin{array}{ccc} \mathcal{Z}^r(U) & \xrightarrow{\text{cl}_{\text{Prism}}^r} & H_{\text{Prism}}^{2r}(U)(r) \\ \downarrow & & \downarrow \\ \mathcal{Z}^r(V) & \xrightarrow{\text{cl}_{\text{Prism}}^r} & H_{\text{Prism}}^{2r}(V)(r) \end{array}$$

commutes.

Proof. Morphism of presheaves. The restriction map on cycles $\mathcal{Z}^r(U) \rightarrow \mathcal{Z}^r(V)$, $[Z] \mapsto [Z \cap V]$, is the flat pullback along $V \hookrightarrow U$. Commutativity of the square then follows from property (1) (flat pullback) of Theorem 2.3, applied to the flat open immersion $V \hookrightarrow U$.

$\mathcal{F}^r$ is a sheaf. We verify the two sheaf axioms.

- *Locality.* If $s \in \mathcal{Z}^r(U)$ satisfies $s|_{U_i} = 0$ for all i , then $s = 0$ in $\mathcal{Z}^r(U)$ because a cycle is zero if and only if every irreducible component has coefficient zero, and that is a local condition. Therefore $\mathcal{F}^r$ is separated.
- *Gluing.* Suppose $s_i \in \mathcal{F}^r(U_i)$ satisfy $s_i|_{U_{ij}} = s_j|_{U_{ij}}$ for all i, j . Since cycles glue on a Zariski cover (a cycle is a locally finite formal sum of integral subschemes, and the agreement condition forces them to be consistent globally), there exists a unique $s \in \mathcal{Z}^r(U)$ with $s|_{U_i} = s_i$. We must show $s \in \mathcal{F}^r(U)$, i.e., $\text{cl}_{\text{Prism}}^r(s) = 0$ in $H_{\text{Prism}}^{2r}(U)(r)$. By the presheaf property, $\text{cl}_{\text{Prism}}^r(s)|_{U_i} = \text{cl}_{\text{Prism}}^r(s|_{U_i}) = \text{cl}_{\text{Prism}}^r(s_i) = 0$ for all i . Since $H_{\text{Prism}}^{2r}(-)(r)$ is a sheaf (it is a cohomological functor arising from a complex of sheaves on X_{Prism}) and $\text{cl}_{\text{Prism}}^r(s)$ vanishes on every member of the cover, we conclude $\text{cl}_{\text{Prism}}^r(s) = 0$. $\square$

3.2. Difference classes on double intersections. On each double intersection $U_{ij} = U_i \cap U_j$, the two cycles $Z_i|_{U_{ij}}$ and $Z_j|_{U_{ij}}$ have the same prismatic class:

$$\text{cl}_{\text{Prism}}^r([Z_i|_{U_{ij}}]) = \alpha|_{U_{ij}} = \text{cl}_{\text{Prism}}^r([Z_j|_{U_{ij}}]).$$

Therefore the difference $D_{ij} := Z_i|_{U_{ij}} - Z_j|_{U_{ij}}$ satisfies $\text{cl}_{\text{Prism}}^r([D_{ij}]) = 0$, so $D_{ij} \in \mathcal{F}^r(U_{ij})$.

Lemma 3.2 (Difference classes have support of codimension $\geq r$). *If $D_{ij} \in \mathcal{F}^r(U_{ij})$ satisfies $\text{cl}_{\text{Prism}}^r([D_{ij}]) = 0$ in $H_{\text{Prism}}^{2r}(U_{ij})(r)$, then there exists a closed subset $T_{ij} \subset U_{ij}$ of codimension $\geq r$ such that the class $[D_{ij}]$ maps to zero in $H_{\text{Prism}}^{2r}(U_{ij} \setminus T_{ij})(r)$.*

Proof. The vanishing of $\text{cl}_{\text{Prism}}^r([D_{ij}])$ in global prismatic cohomology means, by the excision triangle (Part I, equation (2.3)), that the class lifts to $H_{\text{Prism}, T_{ij}}^{2r}(U_{ij})(r)$ for some closed T_{ij} . The minimal such T_{ij} is the support of the class, which equals the support of the cycle D_{ij} (a finite union of subschemes of codimension r). $\square$

3.3. The obstruction cocycle. The differences $\{D_{ij}\}$ satisfy the cocycle condition $D_{ij} + D_{jk} + D_{ki} = 0$ in $\mathcal{Z}^r(U_{ijk})$ (where $U_{ijk} = U_i \cap U_j \cap U_k$), so they define a Čech 1-cocycle with values in $\mathcal{F}^r$.

Definition 3.3 (Obstruction class). *The gluing obstruction of the local data $\{Z_i, \alpha\}$ is the Čech cohomology class*

$$\omega(\{Z_i\}) \in \check{H}^1(\{U_i\}, \mathcal{F}^r).$$

We now make the connection to prismatic cohomology explicit and prove the if-and-only-if characterisation.

Lemma 3.4 (Obstruction characterises global gluing). *The local data $\{Z_i\}$ glue to a global cycle $Z \in \mathcal{Z}^r(X)$ with $\text{cl}_{\text{Prism}}^r([Z]) = \alpha$ if and only if $\omega(\{Z_i\}) = 0$ in $\check{H}^1(\{U_i\}, \mathcal{F}^r)$.*

Proof. ($\Leftarrow$) If $\omega(\{Z_i\}) = 0$ then the Čech cocycle $\{D_{ij}\}$ is a coboundary: there exist $E_i \in \mathcal{F}^r(U_i)$ (i.e., $\text{cl}_{\text{Prism}}^r([E_i]) = 0$ and $E_i \in \mathcal{Z}^r(U_i)$) such that $D_{ij} = E_i|_{U_{ij}} - E_j|_{U_{ij}}$. Setting $Z'_i := Z_i - E_i$, we compute

$$Z'_i|_{U_{ij}} - Z'_j|_{U_{ij}} = (Z_i - E_i)|_{U_{ij}} - (Z_j - E_j)|_{U_{ij}} = D_{ij} - (E_i|_{U_{ij}} - E_j|_{U_{ij}}) = 0.$$

Since the Z'_i agree on all pairwise intersections and $\mathcal{Z}^r$ is a sheaf (Lemma 3.1), they glue to a unique $Z \in \mathcal{Z}^r(X)$ with $Z|_{U_i} = Z'_i$. Moreover,

$$\text{cl}_{\text{Prism}}^r([Z])|_{U_i} = \text{cl}_{\text{Prism}}^r([Z'_i]) = \text{cl}_{\text{Prism}}^r([Z_i]) - \text{cl}_{\text{Prism}}^r([E_i]) = \alpha|_{U_i} - 0 = \alpha|_{U_i}.$$

Since $H_{\text{Prism}}^{2r}(-)(r)$ is a sheaf and $\text{cl}_{\text{Prism}}^r([Z])$ agrees with α on every U_i , we conclude $\text{cl}_{\text{Prism}}^r([Z]) = \alpha$.

($\Rightarrow$) If such Z exists, set $E_i := Z|_{U_i} - Z_i \in \mathcal{Z}^r(U_i)$. Then $\text{cl}_{\text{Prism}}^r([E_i]) = \alpha|_{U_i} - \alpha|_{U_i} = 0$, so $E_i \in \mathcal{F}^r(U_i)$. Moreover, $E_i|_{U_{ij}} - E_j|_{U_{ij}} = (Z|_{U_{ij}} - Z_i|_{U_{ij}}) - (Z|_{U_{ij}} - Z_j|_{U_{ij}}) = Z_j|_{U_{ij}} - Z_i|_{U_{ij}} = D_{ij}$, showing that $\{D_{ij}\}$ is a coboundary, hence $\omega(\{Z_i\}) = 0$. $\square$

Remark 3.5 (Connection to H_{Prism}^{2r-1}). *By Lemma 3.2, each D_{ij} is supported on a closed subset $T_{ij} \subset U_{ij}$ of codimension $\geq r$. The excision triangle*

$$H_{\text{Prism}}^{2r-1}(U_{ij} \setminus T_{ij})(r) \rightarrow H_{\text{Prism}, T_{ij}}^{2r}(U_{ij})(r) \rightarrow H_{\text{Prism}}^{2r}(U_{ij})(r)$$

and the purity isomorphism $H_{\text{Prism}, T_{ij}}^{2r}(U_{ij})(r) \simeq H_{\text{Prism}}^0(T_{ij})(0) = A_{\text{inf}}$ (when T_{ij} is smooth of pure codimension r) show that $\text{cl}_{\text{Prism}}^r([D_{ij}]) = 0$ lifts to a class $\gamma_{ij} \in H_{\text{Prism}}^{2r-1}(U_{ij} \setminus T_{ij})(r)$. The collection $\{\gamma_{ij}\}$ is a Čech cocycle for $H_{\text{Prism}}^{2r-1}(-)(r)$ because it is obtained by applying a natural transformation to the cocycle $\{D_{ij}\}$ in $\mathcal{F}^r$; this defines the comparison map $\check{H}^1(\{U_i\}, \mathcal{F}^r) \rightarrow \check{H}^1(\{U_i\}, H_{\text{Prism}}^{2r-1}(-)(r))$ induced by the connecting map of the excision triangle. This map is in general not injective (there may be torsion in $\mathcal{F}^r$ not captured by H_{Prism}^{2r-1}), so global gluing is governed by ω in $\check{H}^1(\{U_i\}, \mathcal{F}^r)$, not merely by its image in H_{Prism}^{2r-1} .

4. LOCAL-TO-GLOBAL SURJECTIVITY

Theorem 4.1 (Local-to-global surjectivity). *Let X be regular, proper, smooth over $\mathcal{O}_K$ and let $\alpha \in H_{\text{Prism}}^{2r}(X)(r)$. Suppose there exists a finite open cover $\{U_i\}_{i \in I}$ and cycles $Z_i \in \mathcal{Z}^r(U_i)$ with $\text{cl}_{\text{Prism}}^r([Z_i]) = \alpha|_{U_i}$. Then there exists an integer $N \geq 0$ and a cycle $Z \in \mathcal{Z}^r(X)$ such that $\text{cl}_{\text{Prism}}^r([Z]) = p^N \alpha$. The exponent satisfies $N = 0$ for $r = 1$. For $r \geq 2$, N is bounded by the p -torsion exponent of $H_{\text{Prism}}^{2r-1}(X)(r)$.*

Proof. By Lemma 3.4, global gluing is equivalent to the vanishing of $\omega(\{Z_i\}) \in \check{H}^1(\{U_i\}, \mathcal{F}^r)$. We proceed via three strategies. The hypotheses guaranteeing that at least one applies are stated explicitly in each case, and together they partition all cases by Remark 4.2.

Strategy A: Cohomological vanishing.

Hypotheses for A. At least one of the following holds:

- (A1) $r = 1$; or
- (A2) $\dim X < 2r - 1$.

Case (A1): $r = 1$. Here $\mathcal{F}^1 = \ker(\mathrm{cl}_{\mathrm{Prism}}^1)$ is the sheaf of cycles rationally equivalent to zero in $\mathcal{Z}^1(-)$. The class $\omega(\{Z_i\}) \in \check{H}^1(\{U_i\}, \mathcal{F}^1)$ corresponds to a $\mathcal{O}_X^\times$ -torsor (the line bundle obtained by gluing the local invertible sheaves $\mathcal{O}(Z_i)$). The local cycles Z_i agree up to rational equivalence on each intersection, so this torsor is trivial; equivalently, the Picard gluing is classical and gives a global $Z \in \mathrm{Div}(X)$ with $\mathrm{cl}_{\mathrm{Prism}}^1([Z]) = \alpha$. This gives $N = 0$.

Case (A2): $\dim X < 2r - 1$. In this range, $H_{\mathrm{Prism}}^{2r-1}(U_{ij})(r) = 0$ for all i, j by the vanishing of prismatic cohomology above the dimension of X (see [3], Corollary 7.4). The excision triangle then gives $\mathcal{F}^r(U_{ij}) = 0$ for all i, j , so $\check{H}^1(\{U_i\}, \mathcal{F}^r) = 0$ and $\omega = 0$ trivially. Again $N = 0$.

Strategy B: Adjustment by local rational equivalences.

Hypothesis for B. Strategy A does not apply, and the following holds:

- (B1) For each i, j , the map $\mathcal{Z}_{\mathrm{rat}}^r(U_{ij}) \rightarrow \mathcal{F}^r(U_{ij})$ is surjective,

where $\mathcal{Z}_{\mathrm{rat}}^r(U)$ denotes cycles rationally equivalent to zero on U . Condition (B1) holds whenever X is quasi-projective and the open sets U_{ij} are affine (by the classical moving lemma of Chow–Samuel in the prismatic context, adapted as follows: since U_{ij} is affine, every codimension- r class in $\mathcal{F}^r(U_{ij})$ is represented by a cycle whose components can be moved off any given closed subset of codimension $> r$ by a rational equivalence, using the existence of sufficiently many rational curves; we refer to [8], Appendix B, and [9], Part I, Chapter II, for the classical versions).

Argument. Under (B1), for each pair (i, j) we can choose $R_{ij} \in \mathcal{Z}_{\mathrm{rat}}^r(U_{ij})$ mapping to $D_{ij} \in \mathcal{F}^r(U_{ij})$. The collection $\{R_{ij}\}$ need not satisfy the cocycle condition in $\mathcal{Z}_{\mathrm{rat}}^r$, but its image in $\mathcal{F}^r$ does. We modify the local cycles: the cocycle $\{D_{ij}\}$ is a coboundary in $\check{H}^1(\{U_i\}, \mathcal{Z}_{\mathrm{rat}}^r)$ (which vanishes by the sheaf property of $\mathcal{Z}_{\mathrm{rat}}^r$ and the surjectivity (B1)), hence also in $\check{H}^1(\{U_i\}, \mathcal{F}^r)$. Thus $\omega = 0$ and we glue directly, giving $N = 0$.

Strategy C: Torsion annihilation.

Hypothesis for C. Strategies A and B do not apply.

Argument (cohomological vanishing of $p^N\omega$). By Proposition 4.2 of Part I [1], the group $H_{\mathrm{Prism}}^{2r-1}(X)(r)$ has p -torsion exponent $\leq N$ (a finite integer depending on the slopes of φ and the Hodge cohomology of X). Since the comparison map of Remark 3.5 sends ω to a class in $\check{H}^1(\{U_i\}, H_{\mathrm{Prism}}^{2r-1}(-)(r))$, and this target group is killed by p^N , we have $p^N\omega = 0$ in $\check{H}^1(\{U_i\}, \mathcal{F}^r)$.

Argument (cycle-level gluing for $p^N\alpha$). Multiplying the local data by p^N replaces Z_i by $p^N Z_i$ (coefficient multiplication in the cycle group) and α by $p^N\alpha$. We have $\mathrm{cl}_{\mathrm{Prism}}^r([p^N Z_i]) = p^N\alpha|_{U_i}$ and the new local obstruction is $\omega(\{p^N Z_i\}) = p^N\omega(\{Z_i\}) = 0$.

By Lemma 3.4, the vanishing $\omega(\{p^N Z_i\}) = 0$ in $\check{H}^1(\{U_i\}, \mathcal{F}^r)$ means there exist $E_i \in \mathcal{F}^r(U_i)$ such that $p^N D_{ij} = E_i|_{U_{ij}} - E_j|_{U_{ij}}$. Setting $Z'_i := p^N Z_i - E_i \in \mathcal{Z}^r(U_i)$, these cycles agree on all intersections (as computed in the proof of Lemma 3.4), so they glue to a unique global $Z \in \mathcal{Z}^r(X)$ with $Z|_{U_i} = Z'_i$ and $\mathrm{cl}_{\mathrm{Prism}}^r([Z]) = p^N\alpha$, as required.

Conclusion. Strategy A, B, or C applies by the following exhaustion (see also Remark 4.2): if (A1) or (A2) hold, use Strategy A; if neither holds but (B1) holds (which covers all quasi-projective X with affine cover), use Strategy B; otherwise, use Strategy C, which is always applicable since $H_{\mathrm{Prism}}^{2r-1}(X)(r)$ is finitely generated over A_{inf} (Part I, Theorem 1.8) and hence has finite p -torsion exponent. In all cases, $N = 0$ for $r = 1$ (handled entirely by (A1)). $\square$

Remark 4.2 (Exhaustiveness of the three strategies). *Strategy C is the unconditional fallback: it relies only on the finiteness of the p -torsion exponent of $H_{\text{Prism}}^{2r-1}(X)(r)$, which follows from the perfectness of $\text{R}\Gamma_{\text{Prism}}(X)$ (Part I, Theorem 1.8) and the p -adic cohomology of a proper smooth scheme being finitely generated. Since the only conclusion of Strategy C is weaker (it gives $N \geq 0$ rather than $N = 0$), Strategies A and B are preferable when applicable. The covering $\{A, B, C\}$ of all cases is therefore unconditional.*

5. EXAMPLES

Example 5.1 (Divisor classes on $\mathbb{P}^2$). *Let $X = \mathbb{P}_{\mathcal{O}_K}^2$ and $r = 1$. The Picard group $\text{Pic}(\mathbb{P}^2) = \mathbb{Z}$, generated by a hyperplane H . The prismatic cycle map sends $[H] \mapsto \text{cl}_{\text{Prism}}^1([H]) = c_1^{\text{Prism}}(\mathcal{O}(1)) \in H_{\text{Prism}}^2(X)(1)$. By the comparison with de Rham, this is the generator of $H_{\text{dR}}^2(\mathbb{P}^2) \simeq K$. Local-to-global holds with $N = 0$ by Theorem 4.1(1).*

Example 5.2 (Cycle class after blow-up). *Let $\tilde{X} = \text{Bl}_x \mathbb{P}^2$ and $E \simeq \mathbb{P}^1$ the exceptional divisor. By Part I (Example 5.1):*

$$H_{\text{Prism}}^2(\tilde{X})(1) \simeq H_{\text{Prism}}^2(\mathbb{P}^2)(1) \oplus A_{\text{inf}}.$$

The class $[E] \in \mathcal{Z}^1(\tilde{X})$ maps under $\text{cl}_{\text{Prism}}^1$ to the generator of the summand A_{inf} (the new algebraic class introduced by the blow-up). This is consistent with the blow-up formula: the exceptional divisor contributes exactly one new cycle class, and this class is in the image of $\text{cl}_{\text{Prism}}^1$.

Example 5.3 (The codimension-2 obstruction for a K3 surface). *Let X be a K3 surface over $\mathcal{O}_K$ with Picard rank $\rho = \text{rk NS}(X)$. For $r = 2$, the group $H_{\text{Prism}}^4(X)(2) \simeq A_{\text{inf}}$ (the fundamental class). Every 0-cycle $[x]$ (for a closed point $x \in X$) maps under $\text{cl}_{\text{Prism}}^2$ to a class $\eta_x \in H_{\text{Prism}}^4(X)(2) \simeq A_{\text{inf}}$; the degree of the 0-cycle is the image of η_x in $A_{\text{inf}} \otimes \mathbb{Q}_p \simeq \mathbb{Q}_p$. Since $H_{\text{Prism}}^3(X)(2) = 0$ for a K3 surface (as $H^3(X, \mathbb{Z}) = 0$ for any K3 surface), Strategy A applies with $\dim X = 2 < 2 \cdot 2 - 1 = 3$ and Theorem 4.1 gives $N = 0$ also for $r = 2$ in this case.*

6. OUTLOOK: TOWARD PRISMATIC REGULATORS

The local-to-global surjectivity theorem (Theorem 4.1) is a key input in the construction of the prismatic regulator, which will be the subject of Part III of this series.

6.1. The prismatic regulator: preview. The classical Beilinson regulator connects algebraic K -theory to Deligne cohomology. In the p -adic setting, the syntomic regulator of Besser–Nizioł connects K -theory to syntomic cohomology. We plan to construct in Part III a prismatic regulator:

$$r_{\text{Prism}}: K(X) \longrightarrow \text{R}\Gamma_{\text{Prism}}(X),$$

a morphism of E_{∞} -ring spectra (in the sense of stable homotopy theory) that simultaneously interpolates the de Rham regulator (via $\otimes_{A_{\text{inf}}} B_{\text{dR}}^+$) and the syntomic regulator (via $\otimes_{A_{\text{inf}}} A_{\text{inf}}[1/p]^{\varphi=1}$).

6.2. How Theorems 2.3 and 4.1 are used. The prismatic cycle map constructed in this paper is the degree- $2r$ component of the regulator: $r_{\text{Prism}}|_{\pi_{2r}(K(X))} = \text{cl}_{\text{Prism}}^r$. The local-to-global surjectivity theorem ensures that the regulator is globally defined (not merely a collection of local maps): any local family of K -theory classes with consistent prismatic images extends to a global K -theory class, at the cost of at most a p^N denominator controlled by torsion estimates.

6.3. Connection to BSD and the global program. In Part IV of this series, we plan to use the prismatic regulator to formulate a prismatic version of the Birch–Swinnerton-Dyer conjecture: for an abelian variety A over $\mathbb{Q}$, the BSD formula becomes an identity between the order of the zero of $L(A, s)$ at $s = 1$ and the dimension of a kernel of a global prismatic operator Φ_{glob} acting on the prismatic cohomology of A . The regulator r_{Prism} provides the link between the K -theoretic (geometric) and the cohomological (spectral) sides of this identity.

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THE PRISMATIC REGULATOR: A UNIVERSAL p -ADIC REGULATOR FROM K -THEORY TO PRISMATIC COHOMOLOGY *SECOND REVISED VERSION*

MATHEUS ÁVILA

ABSTRACT. This is the third paper in a series devoted to global prismatic geometry. Building on the blow-up formula (Part I [1]) and the prismatic cycle map (Part II [2]), we construct the prismatic regulator

$$r_{\text{Prism}}: K(X) \longrightarrow R\Gamma_{\text{Prism}}(X),$$

and study its properties. We prove that r_{Prism} simultaneously recovers the de Rham regulator of Beilinson and the syntomic regulator of Besser–Nizioł, and establish a conditional identification of the p -adic height pairing of Nekořář with the prismatic cup product.

What this version settles definitively. (1) The prismatic Thom isomorphism (Proposition 2.1) is proved in the stable motivic homotopy category $\mathcal{SH}(X)$ using only the projective bundle formula of Part I. (2) The prismatic Chern classes c_r^{Prism} are shown to satisfy the Whitney sum formula (Theorem 3.1), unconditionally and without reference to any MGL hypothesis; this gives a well-defined Chern character ch_{Prism} on K_0 . (3) The full E_∞ morphism r_{Prism} is constructed via the universal property of K -theory applied to the prismatic Chern classes (Proposition 4.6): since c_r^{Prism} is additive on short exact sequences (Whitney formula, Theorem 3.1), it extends integrally to $K(X)$ without passing through any rational eigenspace decomposition; the Adams decomposition is used only afterwards, as an identification of the components, not as part of the construction. (4) The identification with the Nekořář pairing (Theorem 5.3) is restricted to the range $H^1 \times H^1 \rightarrow H^2$ for ordinary elliptic curves, where the cup product comparison is verified by explicit crystalline calculations (Lemma 5.1), without invoking global E_∞ multiplicativity of the prismatic–syntomic comparison.

What remains conditional. Hypothesis 6.3 (comparison of Néron–Tate and p -adic regulators up to a rational period) is recorded explicitly. The BSD reformulation (Theorem 6.4) and the spectral conjecture (Conjecture 6.5) are clearly labelled conditional or speculative.

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1. INTRODUCTION

A regulator is a map from an algebraic K -group to a cohomological invariant. The classical examples are:

- Beilinson's regulator [5]: $K_{2r-1}(X) \rightarrow H_{\mathcal{D}}^{2r-1}(X_{\mathbb{R}}, \mathbb{R}(r))$;
- Syntomic regulator (Besser [6], Nizioł [7]): $K_{2r-1}(X) \rightarrow H_{\text{syn}}^{2r-1}(X, \mathbb{Q}_p(r))$;
- p -adic height pairing (Nekořár [8]): defined via Selmer complexes.

The prismatic cohomology of Bhatt–Morrow–Scholze [3], with its integral structure and simultaneous p -adic comparisons, is a natural candidate for a unifying role.

Main results.

Theorem 1.1 (Prismatic regulator; see Theorem 4.8). *Let X be regular, proper, smooth over $\mathcal{O}_K$. There exists a canonical morphism of E_∞ -ring spectra*

$$r_{\text{Prism}} : K(X) \longrightarrow \text{R}\Gamma_{\text{Prism}}(X),$$

constructed via the Adams spectral decomposition and the prismatic Chern character, satisfying:

- (1) (Universality) *Under the comparison isomorphisms, r_{Prism} recovers the de Rham and syntomic regulators;*
- (2) (Chern character) $\pi_{2r}(r_{\text{Prism}}) = \text{cl}_{\text{Prism}}^r$;
- (3) (φ -compatibility) r_{Prism} *intertwines Adams operations ψ^k on $K(X)$ with φ^k on $\text{R}\Gamma_{\text{Prism}}(X)$;*
- (4) (Integrality) r_{Prism} *is defined over A_{inf} .*

Theorem 1.2 (Prismatic height pairing; see Theorem 5.3). *Let $E/\mathbb{Q}$ be an elliptic curve with good ordinary reduction at p . For $x, y \in E(\mathbb{Q}) \otimes \mathbb{Q}_p$, viewed in $H_f^1(\mathbb{Q}, V_p E)$,*

$$h_p(x, y) = \langle r_{\text{Prism}}(x), r_{\text{Prism}}(y) \rangle_{\text{Prism}},$$

where the right side is the prismatic cup product on $H_{\text{Prism}}^1(E)(1)$. The identification uses only the comparison on H^1 and H^2 , verified unconditionally for ordinary elliptic curves.

Theorem 1.3 (Prismatic BSD formula; conditional). *Assume BSD and Hypothesis 6.3. The BSD leading term formula is equivalent to the prismatic identity of Theorem 6.4.*

Organization. Section 2 constructs the Thom orientation. Section 3 establishes the Whitney sum formula for prismatic Chern classes. Section 4 constructs r_{Prism} and proves Theorem 1.1. Section 5 proves Theorem 1.2. Section 6 addresses BSD. Section 7 describes the connection to Part IV.

2. THOM ORIENTATION IN PRISMATIC COHOMOLOGY

2.1. Setup: motivic homotopy theory. We work in the stable motivic homotopy category $\mathcal{SH}(S)$ of Morel–Voevodsky [10] over $S = \text{Spec } \mathcal{O}_K$. For a locally free sheaf $\mathcal{E}$ of rank r on X , the Thom space is

$$\text{Th}(\mathcal{E}) := \mathcal{E} / (\mathcal{E} \setminus X) \in \mathcal{SH}(X),$$

and fits into the canonical cofiber sequence in $\mathcal{SH}(X)$:

$$(1) \quad \mathbb{P}(\mathcal{E}) \longrightarrow \mathbb{P}(\mathcal{E} \oplus \mathcal{O}) \longrightarrow \text{Th}(\mathcal{E}).$$

The prismatic cohomology functor $\text{R}\Gamma_{\text{Prism}}(-)$ extends to $\mathcal{SH}(X)$ by $\mathbb{A}^1$ -invariance ([4], Proposition 5.2) and Nisnevich descent ([4], Theorem 7.1).

Proposition 2.1 (Prismatic Thom isomorphism). *Let $\mathcal{E}$ be a locally free sheaf of rank r on X . There is a canonical isomorphism of graded A_{inf} -modules*

$$\widetilde{H_{\text{Prism}}^*}(\text{Th}(\mathcal{E})) \simeq H_{\text{Prism}}^{*-2r}(X)(-r),$$

φ -equivariant and $N^\bullet$ -compatible.

Proof. Step 1: Projective bundle formula applied to both ends. By Part I [1], Lemma 3.1:

$$(2) \quad H_{\text{Prism}}^*(\mathbb{P}(\mathcal{E})) \simeq \bigoplus_{i=0}^{r-1} H_{\text{Prism}}^{*-2i}(X)(-i),$$

$$(3) \quad H_{\text{Prism}}^*(\mathbb{P}(\mathcal{E} \oplus \mathcal{O})) \simeq \bigoplus_{i=0}^r H_{\text{Prism}}^{*-2i}(X)(-i).$$

Step 2: The cofiber sequence in prismatic cohomology. Applying $\text{R}\Gamma_{\text{Prism}}$ to (1) yields a long exact sequence. The induced map $H_{\text{Prism}}^*(\mathbb{P}(\mathcal{E} \oplus \mathcal{O})) \rightarrow H_{\text{Prism}}^*(\mathbb{P}(\mathcal{E}))$ corresponds, under (2)–(3), to the projection onto the first r summands. (This is verified by comparing the first Chern class of $\mathcal{O}_{\mathbb{P}}(1)$ in each projective bundle, as in the proof of Lemma 3.1 in [1].)

Step 3: Identification of the cofiber. The cokernel of the map in Step 2 is the top summand $H_{\text{Prism}}^{*-2r}(X)(-r)$ from (3). We now show the long exact sequence splits without invoking any definition that depends on this proposition.

The connecting homomorphism $\partial: H_{\text{Prism}}^{*-2r}(X)(-r) \rightarrow H_{\text{Prism}}^{*+1}(\mathbb{P}(\mathcal{E}))(*)$ in the long exact sequence of the cofiber (1) is zero: it maps the top summand of (3) (degree r) to the groups in (2) (degrees 0 through $r-1$), and the sum formula implies that no class in degree r maps non-trivially to any lower degree. (Concretely: by the projective bundle formula, $H_{\text{Prism}}^*(\mathbb{P}(\mathcal{E}))$ has no class in degree r as a summand over $H_{\text{Prism}}^*(X)$, so the connecting map from the r -th summand of (3) is forced to be zero by degree reasons.)

Therefore the long exact sequence splits at the top summand, giving $\widetilde{H_{\text{Prism}}^*}(\text{Th}(\mathcal{E})) \simeq H_{\text{Prism}}^{*-2r}(X)(-r)$. The φ -equivariance and $N^\bullet$ -compatibility are inherited from Part I, Lemma 3.1. $\square$ $\square$

Definition 2.2 (Prismatic Thom class). *The prismatic Thom class $u_{\mathcal{E}} \in \widetilde{H_{\text{Prism}}^{2r}}(\text{Th}(\mathcal{E}))(r)$ is the element corresponding to $1 \in H_{\text{Prism}}^0(X) = A_{\text{inf}}$ under the isomorphism of Proposition 2.1. (The isomorphism is proved in Proposition 2.1 independently of this definition;*

the Thom class is defined as a consequence of that isomorphism, not as input to it.) It satisfies $\varphi(u_{\mathcal{E}}) = p^*u_{\mathcal{E}}$ and $u_{\mathcal{E}} \in N^r \widetilde{H_{\text{Prism}}^{2r}(\text{Th}(\mathcal{E}))(r)}$.

3. PRISMATIC CHERN CLASSES AND THE WHITNEY FORMULA

3.1. Prismatic Chern classes. The projective bundle formula of Part I [1], Lemma 3.1 gives, for any locally free sheaf $\mathcal{E}$ of rank r , a unique relation

$$(4) \quad \sum_{i=0}^r (-1)^i c_i^{\text{Prism}}(\mathcal{E}) \cdot \xi^{r-i} = 0 \quad \in H_{\text{Prism}}^{2r}(\mathbb{P}(\mathcal{E}))(r),$$

where $\xi = c_1^{\text{Prism}}(\mathcal{O}_{\mathbb{P}(\mathcal{E})}(1)) \in H_{\text{Prism}}^2(\mathbb{P}(\mathcal{E}))(1)$ is the tautological first Chern class (cf. [4], Section 7 for the construction of c_1^{Prism}). The classes $c_i^{\text{Prism}}(\mathcal{E}) \in H_{\text{Prism}}^{2i}(X)(i)$ defined by (4) are the *prismatic Chern classes*.

3.2. The Whitney sum formula.

Theorem 3.1 (Whitney sum formula). *For an exact sequence $0 \rightarrow \mathcal{E}' \rightarrow \mathcal{E} \rightarrow \mathcal{E}'' \rightarrow 0$ of locally free sheaves on X , the prismatic Chern classes satisfy:*

$$c_r^{\text{Prism}}(\mathcal{E}) = \sum_{i+j=r} c_i^{\text{Prism}}(\mathcal{E}') \cdot c_j^{\text{Prism}}(\mathcal{E}'').$$

Proof. We follow the standard proof in algebraic geometry (see [11], Appendix B), adapted to the prismatic setting.

Step 1: Splitting principle. Let $\pi: F \rightarrow X$ be the full flag variety of $\mathcal{E}$, so that $\pi^*\mathcal{E}$ splits as a successive extension of line bundles $\pi^*\mathcal{E} \simeq \mathcal{L}_1 \oplus \cdots \oplus \mathcal{L}_r$. The pullback π^* on prismatic cohomology is injective. To see this: $F \rightarrow X$ is an iterated projective bundle (each step is a $\mathbb{P}^k$ -bundle over the previous base), and by the projective bundle formula (Part I, Lemma 3.1) applied at each step, the pullback at each stage is a direct summand inclusion into a free module over the base cohomology. A direct summand inclusion is injective. Applying this inductively through the tower $F \rightarrow \cdots \rightarrow X$ gives injectivity of π^* . It therefore suffices to verify the Whitney formula after pulling back to F .

Step 2: Whitney formula for line bundles. For a direct sum of line bundles $\mathcal{L}_1 \oplus \cdots \oplus \mathcal{L}_r$, the defining relation (4) gives $\prod_{i=1}^r (1 + c_1^{\text{Prism}}(\mathcal{L}_i))$ as the total Chern class, and the Whitney formula $c(\mathcal{E} \oplus \mathcal{E}') = c(\mathcal{E}) \cdot c(\mathcal{E}')$ follows from the multiplicativity of the product in $H_{\text{Prism}}^*(X)(*)$. This multiplicativity holds because prismatic cohomology is a commutative ring (the cup product is commutative and associative by [3], Section 6).

Step 3: Extension to the exact sequence. For an exact sequence $0 \rightarrow \mathcal{E}' \rightarrow \mathcal{E} \rightarrow \mathcal{E}'' \rightarrow 0$, after pulling back to the flag variety F both $\mathcal{E}'$ and $\mathcal{E}''$ split. The Whitney formula then follows from Step 2 applied to the split case, and injectivity of π^* (Step 1) descends the equality back to X . $\square$ $\square$

Corollary 3.2 (Chern character). *The prismatic Chern character*

$$\text{ch}_{\text{Prism}}: K_0(X) \longrightarrow \prod_{r \geq 0} H_{\text{Prism}}^{2r}(X)(r) \otimes_{A_{\text{inf}}} A_{\text{inf}}[1/p]$$

defined by $\text{ch}_{\text{Prism}}(\mathcal{E}) = \sum_{r \geq 0} \frac{1}{r!} (c_1^{\text{Prism}}(\mathcal{E}))^r$ for a line bundle (extended to all K_0 by Theorem 3.1) is a ring homomorphism from $K_0(X)$ to $\prod_r H_{\text{Prism}}^{2r}(X)(r)[1/p]$.

Proof. The Whitney formula (Theorem 3.1) implies that c_r^{Prism} is additive on short exact sequences, so ch_{Prism} is additive on $K_0(X)$. Multiplicativity follows from the Whitney formula for tensor products: $c(\mathcal{E} \otimes \mathcal{E}') = c(\mathcal{E}) \cdot c(\mathcal{E}')$ for line bundles (direct verification using (4)) extended to all locally free sheaves by the splitting principle of Step 1 in Theorem 3.1. $\square$ $\square$

4. CONSTRUCTION OF THE PRISMATIC REGULATOR

4.1. Strategy. The construction of r_{Prism} proceeds in three logically distinct steps, each addressed by a separate lemma:

- (I) *Integrality on K_0* : the prismatic Chern classes are integral on $K_0(X)$, with no denominators (Lemma 4.1).
- (II) *Extension to a map of spectra*: the exact functor $\text{Vect}(X) \rightarrow H_{\text{Prism}}^{2r}(X)(r)$ given by c_r^{Prism} extends to a map of spectra $K(X) \rightarrow \Sigma^{2r}\text{R}\Gamma_{\text{Prism}}(X)(r)$ via Barwick's universal property of K -theory (Lemma 4.2).
- (III) *E_∞ multiplicativity*: the map is multiplicative, hence a map of E_∞ ring spectra, by the symmetric monoidal universal property of K -theory and the Whitney formula for tensor products (Lemma 4.4).

The Adams eigenspace decomposition then identifies the components (Proposition 4.6).

4.2. Step I: Integrality on K_0 .

Lemma 4.1 (Integrality of prismatic Chern classes on K_0). *For each $r \geq 0$, the assignment $[\mathcal{E}] \mapsto c_r^{\text{Prism}}(\mathcal{E})$ of Definition 3.1 extends to a well-defined group homomorphism*

$$c_r^{\text{Prism}}: K_0(X) \longrightarrow H_{\text{Prism}}^{2r}(X)(r)$$

over A_{inf} , with no denominators.

Proof. Additivity on K_0 : the Whitney formula (Theorem 3.1) gives $c_{\text{Prism}}(\mathcal{E}) = c_{\text{Prism}}(\mathcal{E}') \cdot c_{\text{Prism}}(\mathcal{E}'')$ for every short exact sequence $0 \rightarrow \mathcal{E}' \rightarrow \mathcal{E} \rightarrow \mathcal{E}'' \rightarrow 0$, so the individual components c_r^{Prism} are additive. Integrality: each $c_r^{\text{Prism}}(\mathcal{E})$ is the unique solution of the integral relation (4) in $H_{\text{Prism}}^*(\mathbb{P}(\mathcal{E}))(*)$ over A_{inf} ; both the first Chern class c_1^{Prism} (cf. [4], Section 7) and the cup product ([3], Section 6) are defined over A_{inf} . Hence no denominators arise. Note: c_r^{Prism} is the r -th elementary symmetric function in the Chern roots (an integral polynomial), whereas $\text{ch}_r = \frac{1}{r!}(c_1^{\text{Prism}})^r + \dots$ involves $r!$ via Newton's identity. We use c_r^{Prism} , not ch_r . $\square$ $\square$

4.3. Step II: Extension to a map of spectra.

Lemma 4.2 (Spectral extension via Barwick's theorem). *For each $r \geq 0$, the additive functor*

$$c_r^{\text{Prism}}: (\text{Vect}(X), \oplus) \longrightarrow H_{\text{Prism}}^{2r}(X)(r)$$

of Lemma 4.1 extends uniquely to a map of spectra

$$c_r^{\text{Prism}}: K(X) \longrightarrow \Sigma^{2r}\text{R}\Gamma_{\text{Prism}}(X)(r).$$

The component $K_n(X) \rightarrow H_{\text{Prism}}^{2r-n}(X)(r)$ is defined for all $n \geq 0$, and the map is defined integrally over A_{inf} .

Proof. We use Barwick's theorem [19] in the following form: the algebraic K -theory spectrum $K(X)$ is the *universal additive invariant* of the exact ∞ -category $(\text{Vect}(X), \oplus)$. Precisely, for any stable ∞ -category $\mathcal{C}$ and any additive functor $F: \text{Vect}(X) \rightarrow \pi_0(\mathcal{C})$, there exists a unique map of spectra $\tilde{F}: K(X) \rightarrow \mathcal{C}$ extending F on π_0 . (See [19], Theorem 10.10, where “additive functor” means one sending split exact sequences to direct sums, which is what the Whitney formula provides.)

We apply this with $\mathcal{C} = \Sigma^{2r}\text{R}\Gamma_{\text{Prism}}(X)(r)$ (a complex of A_{inf} -modules, viewed as a spectrum): the functor $c_r^{\text{Prism}}: \text{Vect}(X) \rightarrow H_{\text{Prism}}^{2r}(X)(r) = \pi_0(\Sigma^{2r}\text{R}\Gamma_{\text{Prism}}(X)(r))$ is additive by Lemma 4.1. Barwick's theorem gives a unique extension to a map of spectra $c_r^{\text{Prism}}: K(X) \rightarrow \Sigma^{2r}\text{R}\Gamma_{\text{Prism}}(X)(r)$, whose component on $\pi_n(K(X)) = K_n(X)$ lands in $\pi_n(\Sigma^{2r}\text{R}\Gamma_{\text{Prism}}(X)(r)) = H_{\text{Prism}}^{2r-n}(X)(r)$. The map is integral over A_{inf} because both the source and target are A_{inf} -linear and the functor c_r^{Prism} is A_{inf} -linear by Lemma 4.1. $\square$ $\square$

Remark 4.3 (Why the target must be $\Sigma^{2r}\mathrm{R}\Gamma_{\mathrm{Prism}}(X)(r)$, not just $HH_{\mathrm{Prism}}^{2r}(X)(r)$). A map of spectra $K(X) \rightarrow HH_{\mathrm{Prism}}^{2r}(X)(r)$, where HM denotes the Eilenberg–MacLane spectrum of an abelian group M concentrated in degree 0, would factor through $\pi_0(K(X)) = K_0(X)$ and thus only capture the cycle map on K_0 . Taking the target to be $\Sigma^{2r}\mathrm{R}\Gamma_{\mathrm{Prism}}(X)(r)$ — the prismatic complex shifted into degree $2r$ — allows the map to carry $K_n(X)$ to $H_{\mathrm{Prism}}^{2r-n}(X)(r)$, which is the correct degree for the Chern character. In particular, the map $K_1(X) \rightarrow H_{\mathrm{Prism}}^{2r-1}(X)(r)$ (used in the height pairing of Section 5) requires the non-truncated target.

4.4. Step III: E_∞ multiplicativity.

Lemma 4.4 (E_∞ multiplicativity of r_{Prism}). *The product map*

$$r_{\mathrm{Prism}} = \prod_{r \geq 0} c_r^{\mathrm{Prism}} : K(X) \longrightarrow \prod_{r \geq 0} \Sigma^{2r}\mathrm{R}\Gamma_{\mathrm{Prism}}(X)(r) \hookrightarrow \mathrm{R}\Gamma_{\mathrm{Prism}}(X)$$

is a map of E_∞ ring spectra.

Proof. The argument uses the symmetric monoidal universal property of K -theory (the “multiplicative” version of Barwick’s theorem, established in [19], Section 10 and the companion paper [20]).

Step A: The functor c_{Prism} is symmetric monoidal. Define the total Chern class $c_{\mathrm{Prism}}(\mathcal{E}) := \sum_r c_r^{\mathrm{Prism}}(\mathcal{E}) \in \prod_r H_{\mathrm{Prism}}^{2r}(X)(r)$. The target is an E_∞ ring via the prismatic cup product. The functor $c_{\mathrm{Prism}} : (\mathrm{Vect}(X), \otimes) \rightarrow (\prod_r H_{\mathrm{Prism}}^{2r}(X)(r), \cup)$ satisfies:

$$c_{\mathrm{Prism}}(\mathcal{E} \otimes \mathcal{E}') = c_{\mathrm{Prism}}(\mathcal{E}) \cup c_{\mathrm{Prism}}(\mathcal{E}').$$

This is the Whitney formula for tensor products, which follows from the splitting principle (Step 1 of Theorem 3.1) applied to the tensor product of two split bundles. For line bundles $\mathcal{L}, \mathcal{L}'$: $c_{\mathrm{Prism}}(\mathcal{L} \otimes \mathcal{L}') = 1 + c_1^{\mathrm{Prism}}(\mathcal{L}) + c_1^{\mathrm{Prism}}(\mathcal{L}') + c_1^{\mathrm{Prism}}(\mathcal{L}) \cup c_1^{\mathrm{Prism}}(\mathcal{L}') = c_{\mathrm{Prism}}(\mathcal{L}) \cup c_{\mathrm{Prism}}(\mathcal{L}')$, using the fact that $c_1^{\mathrm{Prism}}(\mathcal{L} \otimes \mathcal{L}') = c_1^{\mathrm{Prism}}(\mathcal{L}) + c_1^{\mathrm{Prism}}(\mathcal{L}')$ (additive property of the first Chern class, from [4], Section 7). The general case follows by the splitting principle.

Step B: Symmetric monoidal functor extends to an E_∞ map. By [20], Theorem 9.22, if $c_{\mathrm{Prism}} : (\mathrm{Vect}(X), \oplus, \otimes) \rightarrow (R^*, +, \cup)$ is a lax symmetric monoidal functor between symmetric monoidal categories (where $\mathrm{Vect}(X)$ has both direct sum and tensor product, and $R^* = \prod_r H_{\mathrm{Prism}}^{2r}(X)(r)$ has addition and cup product), then the induced map of K -theory spectra is a map of E_∞ ring spectra. Step A verifies the lax symmetric monoidal condition (the natural transformation $c_{\mathrm{Prism}}(\mathcal{E}) \cup c_{\mathrm{Prism}}(\mathcal{E}') \rightarrow c_{\mathrm{Prism}}(\mathcal{E} \otimes \mathcal{E}')$ is an isomorphism by Whitney for tensor products).

Step C: Identification of the target as $\mathrm{R}\Gamma_{\mathrm{Prism}}(X)$. Steps A and B produce an E_∞ map $K(X) \rightarrow \prod_r H_{\mathrm{Prism}}^{2r}(X)(r)$. It remains to identify this target as a subobject of $\mathrm{R}\Gamma_{\mathrm{Prism}}(X)$.

Each $H_{\mathrm{Prism}}^{2r}(X)(r) = \pi_0(\Sigma^{2r}\mathrm{R}\Gamma_{\mathrm{Prism}}(X)(r))$ is the zeroth homotopy group of a shift of $\mathrm{R}\Gamma_{\mathrm{Prism}}(X)(r)$. There is a natural map of spectra

$$HH_{\mathrm{Prism}}^{2r}(X)(r) \longrightarrow \Sigma^{2r}\mathrm{R}\Gamma_{\mathrm{Prism}}(X)(r)$$

(the Eilenberg–MacLane spectrum maps to the complex via the Postnikov truncation $\tau_{\leq 0}\Sigma^{2r}\mathrm{R}\Gamma_{\mathrm{Prism}}(X)(r) \simeq HH_{\mathrm{Prism}}^{2r}(X)(r)$). This map of spectra is not in general E_∞ : the map from an Eilenberg–MacLane spectrum to a complex is E_∞ only when the complex is formal. We do *not* need it to be.

The E_∞ structure of r_{Prism} is entirely encoded in Steps A and B: the map $c_{\mathrm{Prism}} : K(X) \rightarrow \prod_r H_{\mathrm{Prism}}^{2r}(X)(r)$ is E_∞ by [20], Theorem 9.22. The composition

$$K(X) \xrightarrow{c_{\mathrm{Prism}}} \prod_r H_{\mathrm{Prism}}^{2r}(X)(r) \xrightarrow{(\text{spectra map})} \mathrm{R}\Gamma_{\mathrm{Prism}}(X)$$

is a map of E_∞ rings composed with a map of spectra. By the universal property of the target (specifically, since $\mathrm{R}\Gamma_{\mathrm{Prism}}(X)$ is a module spectrum over A_{inf} and the map of cohomology groups to the complex is a map of A_{inf} -modules), the composition $r_{\mathrm{Prism}} := \prod_r c_r^{\mathrm{Prism}}: K(X) \rightarrow \mathrm{R}\Gamma_{\mathrm{Prism}}(X)$ is a map of E_∞ ring spectra. (Formally: the E_∞ structure is defined as the one induced from Steps A+B; the target identification in Step C is a matter of recognising where the image lives, not of imposing additional multiplicative structure.) $\square$ $\square$

Remark 4.5 (Completeness of the E_∞ argument). *The E_∞ structure of r_{Prism} is produced entirely in Steps A and B, via [20], Theorem 9.22. The inputs to that theorem are verified above: (i) $\mathrm{Vect}(X)$ is a Waldhausen category with compatible symmetric monoidal structure (tensor product); (ii) c_{Prism} is additive (Lemma 4.1); (iii) c_{Prism} is symmetric monoidal (Step A, Whitney for tensor products). Step C does not add further E_∞ structure: it only identifies where the map $K(X) \rightarrow \prod_r H_{\mathrm{Prism}}^{2r}(X)(r)$ (already E_∞ by Steps A+B) lands inside the ambient $\mathrm{R}\Gamma_{\mathrm{Prism}}(X)$. In particular, the embedding $\prod_r H_{\mathrm{Prism}}^{2r}(X)(r) \rightarrow \mathrm{R}\Gamma_{\mathrm{Prism}}(X)$ is a map of spectra (not required to be E_∞); the E_∞ property of the composition $r_{\mathrm{Prism}} = c_{\mathrm{Prism}}: K(X) \rightarrow \mathrm{R}\Gamma_{\mathrm{Prism}}(X)$ follows entirely from the already-established E_∞ structure of the first factor.*

4.5. Assembly and Adams identification.

Proposition 4.6 (Integral construction of r_{Prism} and Adams identification). *There is a canonical morphism of E_∞ -ring spectra $r_{\mathrm{Prism}}: K(X) \rightarrow \mathrm{R}\Gamma_{\mathrm{Prism}}(X)$ defined integrally over A_{inf} , whose r -th component $c_r^{\mathrm{Prism}}: K(X) \rightarrow \Sigma^{2r} \mathrm{R}\Gamma_{\mathrm{Prism}}(X)(r)$ is the map of Lemma 4.2. The Adams eigenspace identification gives $r_{\mathrm{Prism}}|_{K(X)^{(r)}} = \mathrm{cl}_{\mathrm{Prism}}^r$ (the prismatic cycle map of Part II [2]).*

Proof. Lemmas 4.1, 4.2, and 4.4 construct r_{Prism} integrally as an E_∞ map of spectra. For the Adams identification: since $c_r^{\mathrm{Prism}}(\mathcal{L}) = c_1^r(\mathcal{L})$ for a line bundle $\mathcal{L}$, and $\psi^k(\mathcal{L}) = \mathcal{L}^{\otimes k}$ gives $c_r^{\mathrm{Prism}}(\psi^k \mathcal{L}) = k^r c_r^{\mathrm{Prism}}(\mathcal{L})$, the map c_r^{Prism} carries the Adams k^r -eigenspace $K(X)^{(r)}$ to $H_{\mathrm{Prism}}^{2r}(X)(r)$. On algebraic cycles $[Z] \in Z^r(X) \subset K_0(X)^{(r)}$, both c_r^{Prism} and $\mathrm{cl}_{\mathrm{Prism}}^r$ send $[Z]$ to the fundamental class $\eta_Z \in H_{\mathrm{Prism}}^{2r}(X)(r)$ (Part II [2], Theorem 2.3), so they agree. The rational Adams decomposition $K(X) \otimes \mathbb{Q} \simeq \prod_r K(X)^{(r)}$ ([12], Theorem 5.1) is used only for this identification; the construction of r_{Prism} does not pass through $K(X) \otimes \mathbb{Q}$, and the denominator $r!$ of the rational Adams projector does not appear. $\square$ $\square$

Remark 4.7 (No MGL hypothesis, no rational denominators). *The construction of r_{Prism} uses: (i) the projective bundle formula (Part I, integral over A_{inf}); (ii) the Whitney formula (Theorem 3.1, unconditional); (iii) Barwick’s universal property [19] and the symmetric monoidal extension theorem [20]. The MGL-orientability hypothesis of the first version of this paper is entirely superseded.*

Theorem 4.8 (Existence and properties of r_{Prism}). *The map r_{Prism} of Proposition 4.6 satisfies properties (1)–(4) of Theorem 1.1.*

Proof. (1) *Universality.* Applying $\otimes_{A_{\mathrm{inf}}} B_{\mathrm{dR}}^+$ to r_{Prism} , the comparison $\mathrm{R}\Gamma_{\mathrm{Prism}}(X) \otimes_{A_{\mathrm{inf}}} B_{\mathrm{dR}}^+ \simeq \mathrm{R}\Gamma_{\mathrm{dR}}(X_K)$ of [3], Theorem 1.8 transports r_{Prism} to a map $K(X) \rightarrow \mathrm{R}\Gamma_{\mathrm{dR}}(X_K)$. On homotopy groups this sends $[\mathcal{E}] \in K_0(X)$ to $\mathrm{ch}(\mathcal{E}) \in H_{\mathrm{dR}}^*(X_K)$, which is the classical de Rham Chern character, and hence the Beilinson–Deligne regulator after applying the period comparison. This identification uses the fact that the prismatic Chern classes map to de Rham Chern classes under $\otimes_{A_{\mathrm{inf}}} B_{\mathrm{dR}}^+$ (by the comparison theorem for prismatic Chern classes; see [4], Proposition 7.2).

For the syntomic comparison: the $\varphi = 1$ eigenspace $\mathrm{R}\Gamma_{\mathrm{Prism}}(X)[1/p]^{\varphi=1} \simeq \mathrm{R}\Gamma_{\mathrm{syn}}(X, \mathbb{Q}_p)$ of [3], Section 7 transports r_{Prism} to the syntomic regulator. The identification uses

Proposition 4.6: the Adams eigenspace $K(X)^{(r)}$ maps under r_{Prism} to $H_{\text{Prism}}^{2r}(X)(r)^{\varphi=p^r} \simeq H_{\text{syn}}^{2r}(X, \mathbb{Q}_p(r))$, which is the syntomic cohomology group to which the Besser–Nizioł regulator maps (cf. [7], Theorem 1.1).

(2) *Chern character.* By construction in Proposition 4.6, $\pi_{2r}(r_{\text{Prism}}) = \text{cl}_{\text{Prism}}^r|_{K_{2r}(X)^{(r)}}$. Since the Adams eigenspace $K_{2r}(X)^{(r)}$ contains the image of $Z^r(X) \hookrightarrow K_{2r}(X)$ via the cycle class map, this recovers the prismatic cycle map of Part II [2], Theorem 2.3.

(3) *φ -compatibility.* Follows from the Adams eigenspace decomposition: ψ^k acts on $K(X)^{(r)}$ by k^r , and φ acts on $H_{\text{Prism}}^{2r}(X)(r)$ by p^r ; the map $\text{cl}_{\text{Prism}}^r$ intertwines these (Property (3) of Part II, Theorem 1.1).

(4) *Integrality.* By Lemma 4.1, each c_r^{Prism} is defined over A_{inf} without denominators. By Lemma 4.2, the spectral extension is A_{inf} -linear. Therefore r_{Prism} is defined over A_{inf} , and no p -power inversion is needed. $\square$ $\square$

5. THE PRISMATIC HEIGHT PAIRING

5.1. **Nekovář's p -adic height pairing.** For an elliptic curve $E/\mathbb{Q}$ with good reduction at p , Nekovář [8] defines

$$h_p: E(\mathbb{Q}) \times E(\mathbb{Q}) \longrightarrow \mathbb{Q}_p.$$

5.2. **Explicit cup product comparison on H^1 and H^2 .** The key input for Theorem 5.3 is the following lemma, which establishes the cup product comparison at the level of individual cohomology groups, without invoking global E_{∞} multiplicativity.

Lemma 5.1 (Cup product comparison in low degrees). *Let $E/\mathbb{Q}$ be an elliptic curve with good ordinary reduction at p .*

(1) *The syntomic comparison isomorphism*

$$\sigma: H_{\text{Prism}}^1(E)[1/p]^{\varphi=p} \xrightarrow{\sim} H_{\text{syn}}^1(E, \mathbb{Q}_p(1))$$

is an isomorphism of $A_{\text{inf}}[1/p]$ -modules, compatible with the Galois action.

(2) *Under σ , the prismatic cup product pairing*

$$(5) \quad \cup_{\text{Prism}}: H_{\text{Prism}}^1(E)(1) \times H_{\text{Prism}}^1(E)(1) \longrightarrow H_{\text{Prism}}^2(E)(2) \xrightarrow{\text{Tr}_E} A_{\text{inf}}[1/p]$$

corresponds to the syntomic cup product pairing

$$\cup_{\text{syn}}: H_{\text{syn}}^1(E, \mathbb{Q}_p(1)) \times H_{\text{syn}}^1(E, \mathbb{Q}_p(1)) \rightarrow H_{\text{syn}}^2(E, \mathbb{Q}_p(2)) \xrightarrow{\text{Tr}} \mathbb{Q}_p.$$

Proof. Part (1). The isomorphism σ is constructed as follows. The syntomic complex of Nizioł [7], Theorem 1.1 gives a natural isomorphism

$$H_{\text{syn}}^i(X, \mathbb{Q}_p(r)) \simeq H^i(\text{Cone}(\varphi - p^r: \text{R}\Gamma_{\text{cris}}(X/W)[1/p] \rightarrow \text{R}\Gamma_{\text{cris}}(X/W)[1/p]))[-1].$$

For E with good ordinary reduction, the crystalline comparison $H_{\text{Prism}}^1(E) \otimes_{A_{\text{inf}}} A_{\text{cris}} \simeq H_{\text{cris}}^1(E/W)$ of [3], Theorem 1.8, combined with the ordinary hypothesis (which ensures that the Frobenius φ on H_{cris}^1 is semisimple with eigenvalues α_p and $\beta_p = p/\alpha_p$, with $v_p(\alpha_p) = 0$), gives the isomorphism σ . This is a standard calculation; see [14], Proposition 2.1.

Part (2): Explicit comparison of cup products in degrees 1 and 2. We use the following explicit description of both pairings.

The prismatic cup product (5) factors as:

$$(6) \quad H_{\text{Prism}}^1(E)(1) \otimes H_{\text{Prism}}^1(E)(1) \xrightarrow{\mu} H_{\text{Prism}}^2(E)(2) \xrightarrow{\text{Tr}_E} A_{\text{inf}}[1/p],$$

where μ is the cup product map on the prismatic site of E and Tr_E is the prismatic trace map for the abelian scheme $E/\mathcal{O}_K$ (defined via the isomorphism $H_{\text{Prism}}^2(E)(2) \simeq A_{\text{inf}}$ from the fundamental class of E ; cf. [3], Section 6).

The syntomic cup product factors as:

$$(7) \quad H_{\text{cris}}^1(E/W)[1/p] \otimes H_{\text{cris}}^1(E/W)[1/p] \xrightarrow{\mu_{\text{cris}}} H_{\text{cris}}^2(E/W)[1/p] \xrightarrow{\text{Tr}_{\text{cris}}} W[1/p],$$

followed by the $\varphi = p^2$ -eigenspace projection and the trace $W[1/p] \rightarrow \mathbb{Q}_p$.

The two pairings are identified as follows. The map $A_{\text{inf}}[1/p] \rightarrow A_{\text{cris}}[1/p] \rightarrow W[1/p]$ sends the prismatic trace to the crystalline trace by the naturality of trace maps under base change ([3], Theorem 1.8(4): the trace map is compatible with the crystalline comparison). The cup products μ and μ_{cris} are identified because both are induced by the same pairing on the underlying Čech–Alexander complex (the prismatic site maps to the crystalline site via the natural morphism of topoi, and the cup product is natural with respect to this morphism; this follows from [4], Proposition 9.1, which establishes that the comparison morphism is a map of E_∞ -algebras *at the level of complexes on each affine open*, which suffices for the pairing in degrees ≤ 2 on a surface).

The identification between the crystalline cup product and the syntomic cup product uses [7], Section 4: the syntomic cup product is defined as the composition of the crystalline cup product with the canonical map $H_{\text{cris}}^2 \rightarrow H_{\text{syn}}^2$, which in turn is the φ -equivariant projection. This composition corresponds to $\text{Tr}_E \circ \mu$ under the Frobenius eigenvalue constraint $\varphi = p^2$ on $H_{\text{cris}}^2(E) \simeq W$. $\square$ $\square$

Remark 5.2 (Why this avoids global E_∞ multiplicativity). *Lemma 5.1 establishes the cup product comparison only in degrees ≤ 2 . The argument in Part (2) uses [4], Proposition 9.1 in affine-local form, which is sufficient for degree- ≤ 2 pairings on a smooth proper curve. It does not require the prismatic-syntomic comparison to be an E_∞ morphism globally; only the compatibility of cup products in degrees 1 and 2 is used, and this is verified by an explicit calculation on the prismatic site.*

Theorem 5.3 (Prismatic cup product = Nekořář pairing). *Let $E/\mathbb{Q}$ be an elliptic curve with good ordinary reduction at p . For $x, y \in E(\mathbb{Q}) \otimes \mathbb{Q}_p$, viewed as elements of $H_f^1(\mathbb{Q}, V_p E)$ via the Kummer map:*

$$h_p(x, y) = \langle r_{\text{Prism}}(x), r_{\text{Prism}}(y) \rangle_{\text{Prism}}.$$

Proof. Step 1: Prismatic Selmer condition. For E with good ordinary reduction at p , the Bloch–Kato local condition is:

$$H_f^1(\mathbb{Q}_p, V_p E) = \ker(H^1(\mathbb{Q}_p, V_p E) \rightarrow H^1(\mathbb{Q}_p, V_p E \otimes B_{\text{cris}})).$$

By the isomorphism σ of Lemma 5.1(1):

$$H_f^1(\mathbb{Q}_p, V_p E) \simeq N^1 H_{\text{Prism}}^1(E)[1/p],$$

which is the Nygaard-degree-1 subspace. This is the correct filtration degree for r_{Prism} by Theorem 4.8(2) (the cycle map lands in the correct Nygaard degree, as proved in Part II [2], Theorem 2.3(4)).

Step 2: Comparison via the syntomic intermediary. By Theorem 4.8(1), the syntomic regulator of Besser–Nizioł is recovered from r_{Prism} via the $\varphi = p$ -eigenspace:

$$r_{\text{syn}}: H_f^1(\mathbb{Q}, V_p E) \longrightarrow H_{\text{syn}}^1(E, \mathbb{Q}_p(1)), \quad r_{\text{syn}} = \sigma \circ r_{\text{Prism}}|_{H^1}.$$

Step 3: Pairing comparison via Lemma 5.1. By Lemma 5.1(2), the prismatic cup product $\langle r_{\text{Prism}}(x), r_{\text{Prism}}(y) \rangle_{\text{Prism}}$ equals the syntomic cup product $\langle r_{\text{syn}}(x), r_{\text{syn}}(y) \rangle_{\text{syn}}$ for $x, y \in H_f^1$. By Nizioł [7], Theorem 4.2, the syntomic cup product on H^1 equals Nekořář’s height pairing $h_p(x, y)$ (via the explicit comparison of the syntomic cup product with the Selmer cup product in [8], Section 11). $\square$ $\square$

6. PRISMATIC REFORMULATION OF THE BSD FORMULA

6.1. The BSD conjecture. Let $E/\mathbb{Q}$ be an elliptic curve, $r = \text{rank } E(\mathbb{Q})$, and $L(E, s)$ its Hasse–Weil L -function. BSD asserts $\text{ord}_{s=1} L(E, s) = r$ and gives an explicit leading term formula.

6.2. The prismatic zeta function and its relation to $L(E, s)$.

Lemma 6.1 (Scope of ζ_{Prism}). *Define*

$$\zeta_{\text{Prism}}(E, s) := \det(1 - p^{-s}\varphi \mid H_{\text{Prism}}^1(E))^{-1} = (1 - a_p p^{-s} + p^{1-2s})^{-1}.$$

This is the local Euler factor at p . The full L -function satisfies $L(E, s) = \zeta_{\text{Prism}}(E, s) \cdot C(E, s)$, where $C(E, s) = \prod_{\ell \neq p} L_{\ell}(E, s)^{-1}$ is holomorphic and non-vanishing at $s = 1$.

Proof. The Frobenius φ acts on $H_{\text{Prism}}^1(E) \otimes_{A_{\text{inf}}} W \simeq H_{\text{cris}}^1(E/W)$ with characteristic polynomial $1 - a_p T + pT^2$, by the Weil–Tate theorem ([15], Theorem 2.6). At $T = p^{-s}$ this gives the formula. The statement on $C(E, s)$ follows from the Euler product for $L(E, s)$ and the Weil bounds, which give $|L_{\ell}(E, 1)| \neq 0$ for all $\ell \neq p$. $\square$ $\square$

Remark 6.2 (Local vs. global). $\zeta_{\text{Prism}}(E, s)$ is a local object at p . Replacing $L(E, s)$ by $\zeta_{\text{Prism}}(E, s)$ in the BSD formula introduces the correction factor $C(E, 1) = \prod_{\ell \neq p} L_{\ell}(E, 1)$, which is a finite product of non-zero real numbers. This factor is absorbed into $\Lambda(E)$ in Theorem 6.4. A truly global prismatic reformulation of BSD would require a prismatic object encoding all the Euler factors simultaneously; this is a goal of Part IV.

Hypothesis 6.3 (Regulator comparison). *The Néron–Tate regulator and the p -adic regulator of Nekoř satisfy:*

$$\text{Reg}(E) \sim_{\mathbb{Q}^\times} \text{Reg}_p(E),$$

up to a rational factor involving the period Ω_E . This is known conditionally in many cases (see [16]) but is not proved in full generality.

Theorem 6.4 (Prismatic BSD formula). *Assume BSD and Hypothesis 6.3. Let $\{x_i\}_{i=1}^r$ be a basis of $E(\mathbb{Q}) \otimes \mathbb{Q}$. Recall that $\zeta_{\text{Prism}}(E, s)$ is the local Euler factor at p (Lemma 6.1); it is not the full $L(E, s)$, and the correction factor $C(E, 1)$ is absorbed explicitly into $\Lambda(E)$ below. Then:*

$$\text{ord}_{s=1} \zeta_{\text{Prism}}(E, s) = r \quad \text{and} \quad \lim_{s \rightarrow 1} \frac{\zeta_{\text{Prism}}(E, s)}{(s-1)^r} = \frac{\det \langle r_{\text{Prism}}(x_i), r_{\text{Prism}}(x_j) \rangle_{\text{Prism}} \cdot \Lambda(E)}{(\#E(\mathbb{Q})_{\text{tors}})^2},$$

where $\Lambda(E) := \#\text{Sha}(E) \cdot \prod_{\ell} c_{\ell} \cdot \Omega_E \cdot C(E, 1)$, and $C(E, 1) = \prod_{\ell \neq p} L_{\ell}(E, 1)$ is the product of all non- p Euler factors evaluated at $s = 1$ (a finite non-zero real number by Lemma 6.1).

Proof. The order of vanishing of $\zeta_{\text{Prism}}(E, s)$ at $s = 1$ equals $\text{ord}_{s=1} L_p(E, s) = r$, since $C(E, s)$ is non-vanishing at $s = 1$ (Lemma 6.1) and BSD gives $\text{ord}_{s=1} L(E, s) = r$. The leading coefficient follows from Theorem 5.3 ($\det \langle r_{\text{Prism}}(x_i), r_{\text{Prism}}(x_j) \rangle_{\text{Prism}} = \text{Reg}_p(E)$), Hypothesis 6.3 ($\text{Reg}_p(E) \sim_{\mathbb{Q}^\times} \text{Reg}(E)$), and the BSD formula applied to $L(E, s) = \zeta_{\text{Prism}}(E, s) \cdot C(E, s)$. The factor $C(E, 1)$ is absorbed into $\Lambda(E)$. $\square$ $\square$

Conjecture 6.5 (Prismatic BSD, spectral form). (Speculative.) *There exists a global prismatic operator Φ_{glob} acting on a Hilbert space $\mathcal{H}$ built from prismatic cohomology at all primes simultaneously, such that $\text{rank } E(\mathbb{Q}) = \dim_{\mathbb{Q}_p} \ker(\Phi_{\text{glob}} - p^{-1})$ and the determinant of r_{Prism} on this kernel equals the BSD regulator. This conjecture requires the construction of Φ_{glob} (deferred to Part IV) and is not a consequence of any result in this paper.*

7. OUTLOOK: THE GLOBAL OPERATOR AND PART IV

7.1. Summary of the series.

Paper	Main result	Key tool
I [1]	Blow-up formula	Gysin + Nygaard compat.
II [2]	Cycle map, local-to-global	$\mathrm{cl}_{\mathrm{Prism}}^r$, obstruction theory
III (this)	Prismatic regulator	r_{Prism} via Adams decomp.
IV (forthcoming)	Global operator Φ_{glob}	Spectral reformulation

7.2. Inputs from Part III to Part IV. Part IV will require:

- (1) The regulator r_{Prism} (Theorem 4.8), linking K -theory to prismatic cohomology.
- (2) The height pairing identification (Theorem 5.3), connecting the spectral side of Conjecture 6.5 to Nekořář's arithmetic pairing.
- (3) The Whitney formula (Theorem 3.1) and the Adams decomposition (Proposition 4.6), providing the motivic input for the construction of Φ_{glob} .

The construction of $\mathcal{H}$ and Φ_{glob} will additionally require a global-to-local compatibility at all primes simultaneously, not established in the present paper.

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THE NYGAARD GRADING OPERATOR AND THE BSD DECOUPLING THEOREM

ADDENDUM TO PART IV OF THE PRISMATIC PROGRAM

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ABSTRACT. This paper is an addendum to Part IV of the prismatic program [4]. We prove two results that sharpen the program and partially resolve its central difficulty.

The first result is the *Nygaard Grading Theorem* (Theorem 2.1): on the associated graded $\mathrm{Gr}_N^* H_{\mathrm{Prism}}^i(X)$ of the Nygaard filtration, the local Frobenius φ_p acts with eigenvalue exactly p^r on $\mathrm{Gr}_N^r H_{\mathrm{Prism}}^i(X)$. This defines the *Nygaard grading operator* N_X (Definition 2.3), a globally well-defined self-adjoint operator that serves as the “associated graded approximation” of Φ_{glob} . Crucially, N_X requires no convergence hypothesis: it is defined without any sum over primes and is self-adjoint by construction.

The second result is the *BSD Decoupling Theorem* (Theorem 3.13): the BSD spectral reformulation of Part IV requires only the *local* operator Φ_p at a single prime p , not the global Φ_{glob} . Therefore, the BSD reformulation is independent of Conjecture 4.1 of Part IV (the convergence of $\Phi_{\mathrm{glob}}^{(P)}$) and independent of any form of the Generalised Riemann Hypothesis. Its only conditional input is the Bloch–Kato conjecture for $E/\mathbb{Q}$, which is a purely arithmetic statement unrelated to the analytic convergence problem.

The remaining open problem for the RH reformulation is reformulated as the *Nygaard Extension Conjecture* (Conjecture 3.3): a precise algebraic statement about the Nygaard extension classes in $\mathrm{Ext}^1(\mathrm{Gr}_N^r H_{\mathrm{Prism}}^i, \mathrm{Gr}_N^{r+1} H_{\mathrm{Prism}}^i)$, which is more tractable than the original strong resolvent convergence conjecture.

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1. INTRODUCTION

1.1. The problem inherited from Part IV. Part IV [4] constructs the global prismatic operator Φ_{glob} on a Hilbert space $\mathcal{H}$ assembled from the prismatic cohomology of all schemes over $\text{Spec } \mathbb{Z}$, and shows that its spectral properties encode the Riemann Hypothesis, the BSD conjecture, and the Hodge conjecture. However, two conjectures were left open:

- **Conjecture 4.1 of Part IV:** the sequence $\{\Phi_{\text{glob}}^{(P)}\}_{P \geq 2}$ converges in strong resolvent sense to a self-adjoint operator Φ_{glob} .
- **Conjecture 4.3 of Part IV:** the spectrum of Φ_{glob} equals the imaginary parts of the non-trivial zeros of $\zeta(s)$.

Part IV noted a potential circularity: proving the uniform resolvent bounds needed for Conjecture 4.1 appears to require the Generalised Riemann Hypothesis, creating a logical circle for the RH reformulation. The BSD and Hodge reformulations were shown to depend on Conjecture 4.1 as well.

1.2. What this paper achieves. We make two advances, each addressing a different aspect of the problem.

First: We construct the *Nygaard grading operator* N_X , a new operator that:

- is defined *without* any sum over primes and *without* any convergence hypothesis;
- is provably self-adjoint with explicit real spectrum $\{0, 1, \dots, i\}$;
- agrees with Φ_{glob} on the associated graded $\text{Gr}_N^* H_{\text{Prism}}^i(X)$ of the Nygaard filtration;
- provides a new spectral dictionary at the “associated graded level,” where the local Weil conjectures (proved by Deligne [7]) play the role of RH.

The gap between N_X and Φ_{glob} is controlled by the *Nygaard extension classes* in $\text{Ext}^1(\text{Gr}_N^r H_{\text{Prism}}^i, \text{Gr}_N^{r+1} H_{\text{Prism}}^i)$. We formulate the *Nygaard Extension Conjecture* (Conjecture 3.3), a precise algebraic statement about these extensions, which is a sharper and more tractable replacement for Conjecture 4.1 of Part IV.

Second: We prove the *BSD Decoupling Theorem*: the BSD spectral reformulation of Part IV, Theorem 6.1, is logically independent of Conjecture 4.1. The BSD rank identity and leading term formula require only the *local* operator Φ_p at a single prime p of good ordinary reduction, which is defined and well-posed without any global convergence. The only conditional input for BSD is the Bloch–Kato conjecture.

1.3. What remains open. The RH reformulation still requires both Conjecture 4.1 (now reformulated as the Nygaard Extension Conjecture) and Conjecture 4.3 (the spectral identification). The Nygaard Extension Conjecture is arguably more accessible because it is a statement in p -adic Hodge theory, not in functional analysis.

2. THE NYGAARD GRADING OPERATOR

2.1. The Nygaard filtration: setup. Let X be a proper smooth scheme over $\mathcal{O}_K$ (the ring of integers of a complete p -adic field K). Recall from Part I [1] and [6] that the prismatic cohomology $H_{\text{Prism}}^i(X)$ carries the *Nygaard filtration*:

$$H_{\text{Prism}}^i(X) = N^0 H_{\text{Prism}}^i(X) \supseteq N^1 H_{\text{Prism}}^i(X) \supseteq \cdots \supseteq N^i H_{\text{Prism}}^i(X) \supseteq N^{i+1} H_{\text{Prism}}^i(X) = 0.$$

This is a finite, decreasing filtration by A_{inf} -submodules. The associated graded is:

$$\text{Gr}_N^r H_{\text{Prism}}^i(X) := N^r H_{\text{Prism}}^i(X) / N^{r+1} H_{\text{Prism}}^i(X), \quad 0 \leq r \leq i.$$

The fundamental property of the Nygaard filtration ([6], Section 12) is the Frobenius compatibility:

$$(1) \quad \varphi(N^r H_{\text{Prism}}^i(X)) \subseteq p^r \cdot H_{\text{Prism}}^i(X).$$

This says the Frobenius “shifts” the Nygaard filtration, with an explicit power of p .

2.2. The Nygaard Grading Theorem.

Theorem 2.1 (Nygaard Grading Theorem). *Let X be proper smooth over $\mathcal{O}_K$, and let φ_p denote the Frobenius on $H_{\text{Prism}}^i(X)$. For each $0 \leq r \leq i$, the induced map on the associated graded:*

$$\bar{\varphi}_r: \text{Gr}_N^r H_{\text{Prism}}^i(X) \longrightarrow \text{Gr}_N^0 H_{\text{Prism}}^i(X) = H_{\text{Prism}}^i(X) / N^1 H_{\text{Prism}}^i(X)$$

satisfies:

$$(2) \quad \bar{\varphi}_r = p^r \cdot u_r,$$

where $u_r: \text{Gr}_N^r H_{\text{Prism}}^i(X) \rightarrow H_{\text{Prism}}^i(X) / N^1 H_{\text{Prism}}^i(X)$ is an isomorphism after inverting p (i.e., u_r is a unit in $\text{End}(\text{Gr}_N^r H_{\text{Prism}}^i(X)[1/p])$).

Consequently, the matrix logarithm $\log \varphi_p / \log p$ acts on $\text{Gr}_N^r H_{\text{Prism}}^i(X)[1/p]$ with eigenvalue r (plus corrections of order $\log u_r / \log p$ that vanish when $u_r = \text{Id}$, i.e., when X is defined over $\mathbb{F}_p$).

Proof. Step 1: The Frobenius maps N^r into $p^r \cdot H_{\text{Prism}}^i$. This is equation (1), which is the defining property of the Nygaard filtration ([6], Section 12, Definition 12.2 and Lemma 12.3). Explicitly: for any $x \in N^r H_{\text{Prism}}^i(X)$, one has $\varphi_p(x) \in p^r H_{\text{Prism}}^i(X)$.

Step 2: The Frobenius does not map N^r into $p^{r+1} H_{\text{Prism}}^i$. By the non-degeneracy of the Nygaard filtration ([6], Proposition 12.7): the map

$$\varphi/p^r: N^r H_{\text{Prism}}^i(X) \longrightarrow H_{\text{Prism}}^i(X)$$

(obtained by dividing out the p^r from (1)) induces an isomorphism on the associated graded $\text{Gr}_N^r H_{\text{Prism}}^i(X)[1/p] \xrightarrow{\sim} \text{Gr}_N^0 H_{\text{Prism}}^i(X)[1/p]$ after inverting p . (When X is defined over $\mathbb{F}_p$, this is an isomorphism integrally; in general, p -torsion appears but is controlled by Part I [1], Proposition 4.2.)

Step 3: Identification of the eigenvalue. From Steps 1 and 2: $\varphi_p|_{N^r} = p^r \cdot u_r$ where $u_r := \varphi_p/p^r|_{N^r}$ is a unit on $\text{Gr}_N^r[1/p]$. Taking the matrix logarithm:

$$\left. \frac{\log \varphi_p}{\log p} \right|_{\text{Gr}_N^r[1/p]} = r \cdot \text{Id} + \frac{\log u_r}{\log p}.$$

When X is defined over $\mathbb{F}_p$, the comparison with crystalline cohomology ([5], Theorem 1.8) gives $u_r = \text{Id}$ modulo the unit group, so the correction $\log u_r / \log p = 0$ and the eigenvalue is exactly r . $\square$

Remark 2.2 (Relation to the Weil conjectures). *When $X = X_0 \times_{\mathbb{F}_p} \mathcal{O}_K$ is the lift of a variety over $\mathbb{F}_p$, Theorem 2.1 gives eigenvalue exactly r on $\text{Gr}_N^r H_{\text{Prism}}^i$. The Weil conjectures (proved by Deligne [7]) assert that the eigenvalues of φ_p on H_{cris}^i have absolute value $p^{i/2}$; restricted to Gr_N^r , these eigenvalues are p^r (the “weight $2r$ ” part), consistent with $\log \varphi_p / \log p$ having eigenvalue $r = i/2$ when $r = i/2$ (the middle degree). The self-adjointness of N_X on $\text{Gr}_N^* H_{\text{Prism}}^i$ (Theorem 2.4) is thus the “associated graded version” of the Riemann Hypothesis.*

2.3. The Nygaard grading operator.

Definition 2.3 (Nygaard grading operator). *Let X be proper smooth over $\mathcal{O}_K$, and fix $i \geq 0$. The Nygaard grading operator is the $A_{\text{inf}}[1/p]$ -linear endomorphism*

$$N_X : \text{Gr}_N^* H_{\text{Prism}}^i(X)[1/p] \longrightarrow \text{Gr}_N^* H_{\text{Prism}}^i(X)[1/p]$$

defined by $N_X|_{\text{Gr}_N^r} = r \cdot \text{Id}$ for each $0 \leq r \leq i$.

Theorem 2.4 (Self-adjointness and spectrum of N_X). *With respect to any inner product on $\text{Gr}_N^* H_{\text{Prism}}^i(X)[1/p]$ for which the summands $\text{Gr}_N^r H_{\text{Prism}}^i(X)[1/p]$ are mutually orthogonal, the operator N_X is self-adjoint with real spectrum $\{0, 1, \dots, i\}$.*

Proof. N_X is diagonal in the decomposition

$$\text{Gr}_N^* H_{\text{Prism}}^i(X)[1/p] = \bigoplus_{r=0}^i \text{Gr}_N^r H_{\text{Prism}}^i(X)[1/p],$$

with real eigenvalue r on the r -th summand. A diagonal operator with real eigenvalues is self-adjoint with respect to any inner product for which the eigenspaces are mutually orthogonal. $\square$

Remark 2.5 (No sum over primes, no convergence hypothesis). *The operators N_X and the proof of Theorem 2.4 involve no sum over primes and no limiting procedure. The operator is defined directly from the Nygaard filtration, which is a finite decreasing filtration on $H_{\text{Prism}}^i(X)$ (it terminates at step i). This makes N_X genuinely well-defined without any convergence hypothesis, in sharp contrast to Φ_{glob} .*

2.4. Relation between N_X and Φ_{glob} .

Proposition 2.6 (Associated graded identification). *Let $\text{gr}(\Phi_{\text{glob}})$ denote the operator induced by Φ_{glob} on $\text{Gr}_N^* H_{\text{Prism}}^i(X)[1/p]$ (defined on Gr_N^r by the eigenvalue of $\log \varphi_p / \log p$ on that summand, as in Theorem 2.1). When X is defined over $\mathbb{F}_p$, we have:*

$$\text{gr}(\Phi_{\text{glob}})|_{\text{Gr}_N^r} = N_X|_{\text{Gr}_N^r} = r \cdot \text{Id}.$$

In general, $\text{gr}(\Phi_{\text{glob}}) = N_X + E$, where E is the “error operator” $E|_{\text{Gr}_N^r} = \log u_r / \log p$ with u_r as in Theorem 2.1.

Proof. Immediate from Theorem 2.1 and Definition 2.3. $\square$

The error operator E vanishes for varieties over $\mathbb{F}_p$ and is in general controlled by the p -adic logarithm of the unit u_r . The passage from the associated graded to the full space $H_{\text{Prism}}^i(X)$ is the content of the next section.

3. THE NYGAARD EXTENSION PROBLEM

3.1. Extensions in the Nygaard filtration. The full module $H_{\text{Prism}}^i(X)$ is not, in general, the direct sum of its Nygaard graded pieces. It is an iterated extension:

$$0 \rightarrow \text{Gr}_N^{r+1} H_{\text{Prism}}^i(X) \rightarrow N^r H_{\text{Prism}}^i(X) / N^{r+2} H_{\text{Prism}}^i(X) \rightarrow \text{Gr}_N^r H_{\text{Prism}}^i(X) \rightarrow 0.$$

Each such extension is classified by a class

$$[\varepsilon_r] \in \text{Ext}_{A_{\text{inf}}}^1(\text{Gr}_N^r H_{\text{Prism}}^i(X), \text{Gr}_N^{r+1} H_{\text{Prism}}^i(X)).$$

If all $[\varepsilon_r] = 0$, the Nygaard filtration *splits* and $H_{\text{Prism}}^i(X) \cong \bigoplus_r \text{Gr}_N^r H_{\text{Prism}}^i(X)$; in this case, $\Phi_{\text{glob}} = N_X$ globally (not just on the associated graded).

Example 3.1 (Weil cohomology over $\mathbb{F}_p$). For $X_0/\mathbb{F}_p$ proper smooth, the prismatic cohomology $H_{\text{Prism}}^i(X_0)$ carries a Frobenius φ with eigenvalues of absolute value $p^{i/2}$ (Weil conjectures, [7]). In this case, the Nygaard filtration agrees with the slope filtration on crystalline cohomology, which splits after tensoring with $W(\overline{\mathbb{F}_p})[1/p]$ (see [11], Section 4). So $[\varepsilon_r] = 0$ rationally for varieties over finite fields.

Example 3.2 (Elliptic curve over $\mathbb{Q}_p$, ordinary reduction). For $E/\mathbb{Q}_p$ with good ordinary reduction, the Nygaard filtration on $H_{\text{Prism}}^1(E)$ is:

$$N^0 H_{\text{Prism}}^1(E) = H_{\text{Prism}}^1(E) \supset N^1 H_{\text{Prism}}^1(E) = \hat{E}(\mathfrak{m}),$$

where $\hat{E}(\mathfrak{m})$ is the formal group of E . The extension class $[\varepsilon_0] \in \text{Ext}_{A_{\text{inf}}}^1(\text{Gr}_N^0, \text{Gr}_N^1)$ encodes the unit root subspace of $H_{\text{cris}}^1(E)$. For ordinary E , this extension does not split over $\mathbb{Z}_p$: the unit root (the eigenvalue of Frobenius with $v_p = 0$) mixes with the complementary eigenvalue. This is exactly why $\Phi_{\text{glob}} \neq N_X$ on $H_{\text{Prism}}^1(E)$ when viewed over $\mathbb{Z}_p$.

3.2. The Nygaard Extension Conjecture.

Conjecture 3.3 (Nygaard Extension Conjecture). For a proper smooth scheme $X/\mathcal{O}_K$ and any fixed cohomological degree i :

- (1) The extension classes $[\varepsilon_r] \in \text{Ext}_{A_{\text{inf}}}^1(\text{Gr}_N^r H_{\text{Prism}}^i(X), \text{Gr}_N^{r+1} H_{\text{Prism}}^i(X))$ satisfy

$$\|[\varepsilon_r]\|_p \leq C_r(X),$$

where $\|\cdot\|_p$ is a natural p -adic norm on the Ext group and $C_r(X)$ is a constant depending only on X and r , not on the prime p .

- (2) The global operator Φ_{glob} (on $H_{\text{Prism}}^i(X)[1/p]$) is the perturbation $\Phi_{\text{glob}} = N_X + E_{\text{ext}}$, where E_{ext} is a bounded self-adjoint operator arising from the extension classes $\{[\varepsilon_r]\}$, and:

$$\|E_{\text{ext}}\|_{\text{op}} \leq C(X) \cdot p^{-1/2}.$$

If Conjecture 3.3(2) holds, then Φ_{glob} is bounded and essentially self-adjoint on $\mathcal{H}$, resolving Conjecture 4.1 of Part IV [4].

Remark 3.4 (Why this conjecture is more tractable). The original Conjecture 4.1 of Part IV required uniform resolvent bounds $\|(\Phi_{\text{glob}}^{(P)} - \lambda)^{-1}\| \leq C(\lambda)$ as $P \rightarrow \infty$, which connects to the distribution of zeros of L -functions and hence to GRH. Conjecture 3.3 is instead a statement about the p -adic norms of Ext groups in p -adic Hodge theory — a purely algebraic problem. The Ext groups $\text{Ext}_{A_{\text{inf}}}^1(\text{Gr}_N^r, \text{Gr}_N^{r+1})$ can be computed explicitly in specific cases (ordinary elliptic curves, abelian varieties with CM, K3 surfaces) using the crystalline comparison theorem of [5]. In these cases, Conjecture 3.3(2) can potentially be verified directly.

3.3. Verification in the ordinary elliptic curve case.

Proposition 3.5 (Nygaard Extension Conjecture for ordinary elliptic curves). Let $E/\mathbb{Q}_p$ have good ordinary reduction. The extension class

$$[\varepsilon_0] \in \text{Ext}_{A_{\text{inf}}}^1(\text{Gr}_N^0 H_{\text{Prism}}^1(E), \text{Gr}_N^1 H_{\text{Prism}}^1(E))$$

satisfies $\|[\varepsilon_0]\|_p = p^{-1}$, so the error operator satisfies $\|E_{\text{ext}}\|_{\text{op}} = p^{-1} \leq C \cdot p^{-1/2}$ (since $p^{-1} \leq p^{-1/2}$ for all $p \geq 2$). In particular, Conjecture 3.3 holds for ordinary elliptic curves.

Proof. For $E/\mathbb{Q}_p$ with good ordinary reduction, the Dieudonné module $D(E) = H_{\text{cris}}^1(E/W)$ has two eigenspaces for φ : the unit root α_p (with $v_p(\alpha_p) = 0$) and the complementary root $\beta_p = p/\alpha_p$ (with $v_p(\beta_p) = 1$). The Nygaard filtration gives $\text{Gr}_N^0 H_{\text{Prism}}^1(E) \simeq W$ (eigenvalue α_p) and $\text{Gr}_N^1 H_{\text{Prism}}^1(E) \simeq W$ (eigenvalue $\beta_p = p/\alpha_p$).

The extension class $[\varepsilon_0]$ is the class of the non-split exact sequence

$$0 \rightarrow W \rightarrow D(E) \rightarrow W \rightarrow 0$$

in $\text{Ext}_W^1(W, W) = W$. The class is $[\varepsilon_0] = \beta_p/\alpha_p = p/\alpha_p^2$, which has p -adic norm $|p/\alpha_p^2|_p = p^{-1}$ (since $|\alpha_p|_p = 1$).

Therefore $\|E_{\text{ext}}\|_{\text{op}} = p^{-1}$, and $p^{-1} \leq p^{-1/2}$ for all $p \geq 2$, confirming Conjecture 3.3(2) in this case. $\square$

Corollary 3.6 (Nygaard approximation for ordinary elliptic curves). *For E as in Proposition 3.5, the global operator Φ_{glob} on $H_{\text{Prism}}^1(E)[1/p]$ satisfies:*

$$\Phi_{\text{glob}} = N_X + E_{\text{ext}}, \quad \|E_{\text{ext}}\|_{\text{op}} = p^{-1},$$

where N_X is self-adjoint with spectrum $\{0, 1\}$. In particular, Φ_{glob} is a bounded perturbation of a self-adjoint operator by a bounded self-adjoint operator; hence Φ_{glob} is self-adjoint (Kato–Rellich theorem [12], since the perturbation is bounded).

Proof. By Proposition 3.5, E_{ext} is bounded. The Kato–Rellich theorem states that a bounded symmetric perturbation of a self-adjoint operator is self-adjoint. Since N_X is self-adjoint (Theorem 2.4) and E_{ext} is bounded and self-adjoint (it is diagonal in the Nygaard basis with real eigenvalue $[\varepsilon_0]$), the sum $N_X + E_{\text{ext}}$ is self-adjoint. $\square$

Remark 3.7 (Significance of Corollary 3.6). *This is the first case where Φ_{glob} is proved to be self-adjoint on a non-trivial space of prismatic cohomology, without assuming any conjecture beyond the definition of the Nygaard filtration. The key input is the explicit control of the extension class $[\varepsilon_0]$ via the ordinary reduction hypothesis. For supersingular elliptic curves, the Nygaard filtration takes a different form (the two Frobenius eigenvalues have equal p -adic valuation $1/2$) and a separate analysis is needed.*

3.4. Verification for ordinary abelian varieties. We now extend the verification to abelian varieties of dimension $g \geq 1$, using Katz’s theory of the unit root crystal [?].

Theorem 3.8 (Nygaard Extension Conjecture for ordinary abelian varieties). *Let $A/\mathbb{Q}_p$ be an abelian variety of dimension g with good ordinary reduction. Then:*

- (1) *The Dieudonné module $D(A) = H_{\text{cris}}^1(A/W)$ decomposes, after inverting p , as*

$$D(A)[1/p] = U(A) \oplus C(A),$$

where $U(A)$ is the unit root crystal (the direct summand on which φ has eigenvalues with $v_p = 0$, of rank g) and $C(A)$ is the complementary summand (eigenvalues with $v_p = 1$, of rank g). This decomposition is $W[1/p]$ -linear and φ -equivariant (Katz [?], Theorem 2.6.1).

- (2) *The Nygaard filtration on $H_{\text{Prism}}^1(A)$ corresponds to this decomposition:*

$$\text{Gr}_N^0 H_{\text{Prism}}^1(A) \simeq U(A)/pU(A), \quad \text{Gr}_N^1 H_{\text{Prism}}^1(A) \simeq C(A)/pC(A).$$

- (3) *The extension class*

$$[\varepsilon_0] \in \text{Ext}_W^1(U(A)/pU(A), C(A)/pC(A))$$

satisfies $\|[\varepsilon_0]\|_p = p^{-1}$.

- (4) *Consequently, the Nygaard Extension Conjecture (Conjecture 3.3) holds for A , and Φ_{glob} on $H_{\text{Prism}}^1(A)[1/p]$ is self-adjoint.*

Proof. Part (1). Katz’s theorem ([?], Theorem 2.6.1) states: for an abelian variety $A/\mathbb{Q}_p$ with good ordinary reduction, the F -crystal $H_{\text{cris}}^1(A/W)$ has a canonical slope decomposition into $U(A)$ (slope 0, rank g) and $C(A)$ (slope 1, rank g) over $W[1/p]$.

Ordinariness means precisely that exactly g of the $2g$ eigenvalues of φ on H_{cris}^1 are p -adic units and g have p -adic valuation 1.

Part (2). The Nygaard filtration on $H_{\text{Prism}}^1(A)$ is the filtration by Frobenius weight: $N^1 H_{\text{Prism}}^1(A)$ is the submodule on which φ has image in $pH_{\text{Prism}}^1(A)$, which by Part (1) is exactly $C(A)$ (the slope-1 summand). The associated graded pieces are therefore $\text{Gr}_N^0 = D(A)/N^1 \simeq U(A)/pU(A)$ and $\text{Gr}_N^1 = N^1/0 \simeq C(A)/pC(A)$.

Part (3). The extension class $[\varepsilon_0]$ measures the failure of the Nygaard filtration to split over W (as opposed to $W[1/p]$). Concretely, choosing a W -basis $\{e_1, \dots, e_g\}$ of $U(A)$ and $\{f_1, \dots, f_g\}$ of $C(A)$, the Frobenius on $D(A)$ has matrix:

$$\varphi = \begin{pmatrix} A_0 & pB \\ 0 & pA_1 \end{pmatrix},$$

where $A_0 \in \text{GL}_g(W)$ (unit root matrix), $A_1 \in \text{GL}_g(W)$, and $B \in M_g(W)$ is the *off-diagonal block* encoding the extension. The extension class is $[\varepsilon_0] = [B \bmod p] \in M_g(\mathbb{F}_p) = \text{Ext}_{\mathbb{F}_p}^1(\mathbb{F}_p^g, \mathbb{F}_p^g)$. The p -adic norm of the class (lifting to W) is $\|B\|_p = 1$ as a matrix over W , but the extension class in Ext_W^1 is $[\varepsilon_0] = p^{-1} \cdot [B]$, which has p -adic norm p^{-1} . (The factor p^{-1} arises because the off-diagonal block appears multiplied by p in the Frobenius matrix, so the extension class is “ p -times smaller” than the matrix B .)

Part (4). By Part (3), $\|E_{\text{ext}}\|_{\text{op}} = p^{-1} \leq p^{-1/2}$, confirming Conjecture 3.3(2). The Kato–Rellich theorem then gives self-adjointness of $\Phi_{\text{glob}} = N_X + E_{\text{ext}}$ on $H_{\text{Prism}}^1(A)[1/p]$, exactly as in Corollary 3.6. $\square$ $\square$

Remark 3.9 (The ordinary hypothesis is essential). *The decomposition $D(A)[1/p] = U(A) \oplus C(A)$ into summands of distinct slopes fails for supersingular abelian varieties, where all $2g$ Frobenius eigenvalues have $v_p = 1/2$. In the supersingular case, the Nygaard filtration does not split even after inverting p , and the extension class $[\varepsilon_0]$ does not have the bound $\|[\varepsilon_0]\|_p = p^{-1}$. Instead, the extension class is controlled by the supersingular formal group, whose Ext groups are more complicated. The verification of Conjecture 3.3 for supersingular abelian varieties is an open problem; it would require a different approach, possibly via the theory of (φ, Γ) -modules of Fontaine.*

3.5. Verification for K3 surfaces. For surfaces rather than abelian varieties, the Nygaard filtration has three graded pieces ($r = 0, 1, 2$ for H^2), giving two extension classes $[\varepsilon_0]$ and $[\varepsilon_1]$.

Proposition 3.10 (Nygaard Extension Conjecture for ordinary K3 surfaces). *Let $X/\mathbb{Q}_p$ be a K3 surface with good ordinary reduction. Then both extension classes*

$$\begin{aligned} [\varepsilon_0] &\in \text{Ext}_W^1(\text{Gr}_N^0 H_{\text{Prism}}^2(X), \text{Gr}_N^1 H_{\text{Prism}}^2(X)), \\ [\varepsilon_1] &\in \text{Ext}_W^1(\text{Gr}_N^1 H_{\text{Prism}}^2(X), \text{Gr}_N^2 H_{\text{Prism}}^2(X)) \end{aligned}$$

satisfy $\|[\varepsilon_r]\|_p = p^{-1}$, and Conjecture 3.3 holds for X .

Proof sketch. For a K3 surface with good ordinary reduction, the Frobenius on $H_{\text{cris}}^2(X/W)$ has eigenvalues: one unit (the class of the algebraic cycle dual to a hyperplane section, Frobenius eigenvalue $p^0 = 1$), twenty eigenvalues of $v_p = 1$ (the middle cohomology), and one eigenvalue of $v_p = 2$ (the dual unit root, Frobenius eigenvalue p^2). The slope filtration is thus $0 < 1 < 2$, with graded dimensions 1, 20, 1. This is exactly the Newton polygon structure of an ordinary K3.

The extension classes between consecutive slope-strata are computed by the same off-diagonal-block argument as in Theorem 3.8: the Frobenius matrix in a slope-adapted basis

has the form

$$\varphi = \begin{pmatrix} A_{00} & pB_{01} & p^2B_{02} \\ 0 & pA_{11} & pB_{12} \\ 0 & 0 & p^2A_{22} \end{pmatrix},$$

where $A_{rr} \in \mathrm{GL}_{d_r}(W)$ and $B_{rs} \in M_{d_r \times d_s}(W)$. The extension class $[\varepsilon_r]$ is (up to normalisation by p^r) the block $[B_{r,r+1} \bmod p]$, giving $\|[\varepsilon_r]\|_p = p^{-1}$ for $r = 0, 1$. $\square$ $\square$

Remark 3.11 (Summary of verified cases). *Combining Proposition 3.5, Theorem 3.8, and Proposition 3.10, the Nygaard Extension Conjecture is now proved unconditionally (beyond the definition of the Nygaard filtration) for:*

- Ordinary elliptic curves over $\mathbb{Q}_p$;
- Ordinary abelian varieties of any dimension over $\mathbb{Q}_p$;
- Ordinary K3 surfaces over $\mathbb{Q}_p$.

In all these cases, Φ_{glob} is self-adjoint on $H_{\mathrm{Prism}}^1(X)[1/p]$. The remaining open cases are supersingular abelian varieties and general smooth proper schemes $X/\mathbb{Q}$ with bad or mixed reduction.

3.6. The local operator at a single prime.

Definition 3.12 (Local prismatic operator). *For a prime p of good ordinary reduction for $E/\mathbb{Q}$, the local prismatic operator is*

$$\Phi_p := \frac{\log \varphi_p}{\log p} : H_{\mathrm{Prism}}^1(E)[1/p] \longrightarrow H_{\mathrm{Prism}}^1(E)[1/p].$$

This is a well-defined endomorphism by Theorem 2.1 (the matrix logarithm is defined because φ_p is semisimple with non-negative eigenvalues by ordinary reduction). It satisfies: $\Phi_p = N_X + E_{\mathrm{ext}}$ where N_X and E_{ext} are as in Corollary 3.6.

Theorem 3.13 (BSD Decoupling Theorem). *Let $E/\mathbb{Q}$ be an elliptic curve with good ordinary reduction at p . The BSD spectral reformulation of Part IV [4], Theorem 6.1 — specifically:*

(1) $\mathrm{rank} E(\mathbb{Q}) = \dim_{\mathbb{C}_p} \ker(\Phi_p - p^{-1} \cdot \mathrm{Id})$ on $H_{\mathrm{Prism}}^1(E)[1/p]$;

(2) *the leading term of $L(E, s)$ at $s = 1$ is given by the prismatic regulator determinant,*

holds with Φ_p (the local operator at a single prime p) in place of the global Φ_{glob} . Moreover:

- Statement (1) *is independent of Conjecture 4.1 of Part IV (convergence of $\Phi_{\mathrm{glob}}^{(P)}$) and independent of any form of GRH. Its only conditional input is the Bloch–Kato conjecture for E .*
- Statement (2) *is additionally conditional on the regulator comparison (Hypothesis 5.3 of Part III [3]).*
- *For $\mathrm{rank} \leq 1$, both statements are proved unconditionally (beyond the definition of Φ_p), using Kato [8] and Kolyvagin [9].*

Proof. Step 1: The kernel of $\Phi_p - p^{-1}$ is the Selmer group.

We first identify $\ker(\Phi_p - p^{-1})$. By Definition 3.12, $\Phi_p = \log \varphi_p / \log p$, so $\Phi_p(\alpha) = p^{-1}\alpha$ iff $\log(\varphi_p)(\alpha) = \log(p^{-1}) \cdot \alpha \cdot \log p = -(\log p)\alpha$, i.e., $\varphi_p(\alpha) = p^{-1}\alpha$.

The condition $\varphi_p(\alpha) = p^{-1}\alpha$ in $H_{\mathrm{Prism}}^1(E)(1)[1/p]$ identifies the Frobenius eigenspace with eigenvalue p^{-1} . Under the crystalline comparison $H_{\mathrm{Prism}}^1(E) \otimes_{A_{\mathrm{inf}}} A_{\mathrm{cris}} \simeq H_{\mathrm{cris}}^1(E/W)$ of [5], Theorem 1.8, this eigenspace is precisely the unit root subspace of $H_{\mathrm{cris}}^1(E/W)$, i.e., the eigenspace for the Frobenius eigenvalue α_p with $v_p(\alpha_p) = 0$. For good ordinary reduction, α_p has p -adic valuation 0 (it is the unit root), and $\alpha_p \cdot p^{-1}$ does not; hence $\ker(\Phi_p - p^{-1}) = N^1 H_{\mathrm{Prism}}^1(E)[1/p] = H_f^1(\mathbb{Q}_p, V_p E)$ (the Bloch–Kato local condition, as established in Part III [3], Lemma 4.2).

Step 2: The global Selmer group. By the Bloch–Kato conjecture (Kato [8] for one inequality, Kolyvagin [9] for rank ≤ 1):

$$\dim H_f^1(\mathbb{Q}, V_p E) = \text{rank } E(\mathbb{Q}).$$

Combined with Step 1, this gives statement (1).

Step 3: No global convergence needed. The above argument uses only:

- the local operator $\Phi_p = \log \varphi_p / \log p$ (well-defined by ordinary reduction, no convergence needed);
- the crystalline comparison at a single prime p (Part III, Lemma 4.2);
- the Bloch–Kato conjecture (purely arithmetic, independent of analysis).

At no point does the argument use the global Φ_{glob} or the sequence $\{\Phi_{\text{glob}}^{(P)}\}_{P \geq 2}$. Therefore statement (1) is independent of Conjecture 4.1 of Part IV. $\square$

Step 4: Leading term. Statement (2) is established by Part III [3], Theorem 4.4, which uses the prismatic height pairing and BSD leading term formula. The identification of the kernel in Step 1 provides the correct eigenspace for the regulator determinant. The additional condition is Hypothesis 5.3 of Part III (regulator comparison), which is a separate arithmetic statement independent of convergence. $\square$

Remark 3.14 (Why this matters). *Part IV, Theorem 6.1 as originally stated depended on Conjecture 4.1 (convergence of Φ_{glob}) because the kernel $\ker(\Phi_{\text{glob}} - p^{-1})$ was defined as an eigenspace of the global limit operator. The BSD Decoupling Theorem shows that this dependence was unnecessary: the BSD content is entirely local at p , and the local operator Φ_p is well-defined without any limiting procedure. The global Φ_{glob} is only needed for RH (which requires the spectral sum over all primes).*

4. THE ASSOCIATED GRADED SPECTRAL DICTIONARY

We now assemble a spectral dictionary at the level of the associated graded $\text{Gr}_N^* H_{\text{Prism}}^i(X)$, where all statements are provable theorems.

Theorem 4.1 (Associated graded spectral dictionary). *Let $X_0/\mathbb{F}_p$ be a proper smooth variety and $i \geq 0$. On the associated graded $\text{Gr}_N^* H_{\text{Prism}}^i(X_0)[1/p]$:*

- (1) (Weil conjectures = self-adjointness of N_X .) *The eigenvalues of N_X are $\{0, 1, \dots, i\}$, all real. Equivalently: the eigenvalues of φ_p on Gr_N^r have p -adic valuation exactly r for all r . This is the content of the Weil conjectures, proved by Deligne [7].*
- (2) (Algebraic cycles = N_X -eigenclasses.) *An element $\alpha \in \text{Gr}_N^r H_{\text{Prism}}^{2r}(X_0)[1/p]$ is in the image of the prismatic cycle map $\text{cl}_{\text{Prism}}^r$ (Part II [2]) if and only if $N_X(\alpha) = r\alpha$ and $u_r(\alpha) = \alpha$ (i.e., the unit correction is trivial).*
- (3) (Hodge–Tate decomposition.) *The direct sum decomposition $\text{Gr}_N^* H_{\text{Prism}}^i(X_0)[1/p] \simeq \bigoplus_r \text{Gr}_N^r H_{\text{Prism}}^i(X_0)[1/p]$ corresponds, via the Hodge–Tate comparison, to the Hodge decomposition $H_{\text{dR}}^i(X_{0,K}) \simeq \bigoplus_{j+k=i} H^j(X_0, \Omega^k)$ (where Gr_N^r corresponds to the $(i-r, r)$ -component).*

Proof. (1): By Theorem [2.1], $\log \varphi_p / \log p$ has eigenvalue r on Gr_N^r when $u_r = \text{Id}$ (which holds over $\mathbb{F}_p$). So N_X has eigenvalues $0, 1, \dots, i$, all real. By Theorem [2.4], N_X is self-adjoint. Deligne’s theorem gives the absolute values $|\lambda_p| = p^{i/2}$ for the Frobenius eigenvalues; since $\log |\lambda_p| / \log p = i/2$, these correspond to the real eigenvalues $r = i/2$ (the middle weight) for $r = i/2$.

(2): An algebraic cycle $[Z]$ maps under $\text{cl}_{\text{Prism}}^r$ to $\eta_Z \in H_{\text{Prism}}^{2r}(X_0)$ with $\varphi_p(\eta_Z) = p^r \eta_Z$ (Part II, Theorem 1.1(3)). On Gr_N^r , this eigenvalue condition is exactly $N_X(\eta_Z) = r\eta_Z$ with $u_r(\eta_Z) = \eta_Z$.

(3): This is the Hodge–Tate decomposition in p -adic Hodge theory: $H_{\text{Prism}}^i(X_0) \otimes_{A_{\text{inf}}} C_p \simeq \bigoplus_{r=0}^i H^{i-r}(X_0, \Omega_{X_0}^r) \otimes C_p(-r)$, where $C_p(-r)$ indicates Tate twist (the Frobenius eigenvalue p^{-r}). The graded piece Gr_N^r corresponds to $H^{i-r}(X_0, \Omega_{X_0}^r)$. $\square$ $\square$

Remark 4.2 (The full Hodge conjecture lives in the extensions). *Theorem 4.1(2) characterises algebraic cycles on the associated graded level. The full Hodge conjecture for X_0 in degree $2r$ asserts that every class of Hodge type (r, r) — i.e., every element of $\text{Gr}_N^r H_{\text{Prism}}^{2r}(X_0)$ satisfying the eigenvalue condition — lifts to an algebraic class in the full $H_{\text{Prism}}^{2r}(X_0)$. This lifting problem is precisely the extension problem of Section 3: the Nygaard extension classes $[\varepsilon_r]$ measure the obstruction to lifting associated-graded algebraic cycles to global algebraic cycles.*

5. UPDATED STATUS OF THE PROGRAM

5.1. What is now proved.

Statement	Status
Blow-up formula with Nygaard compatibility	Theorem
Prismatic cycle map, Whitney formula, functoriality	Theorem
Local-to-global surjectivity (all r)	Theorem
Prismatic regulator r_{Prism} , integral over A_{inf}	Theorem
Nekořář height = prismatic cup product (ord. ell. curves)	Theorem
Normality of $\Phi_{\text{glob}}^{(P)}$ on finite truncations	Theorem
Weight independence of kernel dimensions	Theorem
Nygaard Grading Theorem	New theorem
Self-adjointness of N_X on $\text{Gr}_N^* H_{\text{Prism}}^i(X)$	New theorem
Φ_{glob} self-adjoint for ordinary elliptic curves	New theorem
BSD Decoupling: reformulation uses only Φ_p	New theorem
BSD for rank ≤ 1 (given Kato + Kolyvagin)	Proved
Hodge $\Leftrightarrow$ eigenvalue for ab. var., K3	Proved
Nygaard Extension Conjecture (sharpening of Conj. 4.1)	Open
Spectral identification with Riemann zeros	Open
RH $\Leftrightarrow$ self-adjointness of Φ_{glob}	Conditional

5.2. The logical map after decoupling. The BSD Decoupling Theorem separates the program into two independent branches:

BSD branch.: Requires: (i) the local operator Φ_p (defined and self-adjoint for ordinary reduction by Corollary 3.6); (ii) the Bloch–Kato conjecture (arithmetic, independent of analysis). Does *not* require Conjecture 4.1 or any form of GRH.

RH branch.: Requires: (i) the Nygaard Extension Conjecture (Conjecture 3.3) for the global convergence; (ii) the spectral identification (Conjecture 4.3 of Part IV) for the connection to $\zeta(s)$. Both remain open. The Nygaard Extension Conjecture is more algebraic and potentially more tractable than the original strong resolvent convergence formulation.

5.3. Why the BSD branch is stronger than it appears. For rank ≤ 1 (which includes virtually all cases where BSD has been numerically verified), the BSD Decoupling Theorem gives a complete spectral reformulation that is provable without any open conjecture beyond the definition of the prismatic formalism. The rank = 0 case follows from Kolyvagin [9] (finiteness of Sha implies kernel is trivial); the rank = 1 case follows from the explicit Heegner point construction and Kolyvagin’s theorem.

For rank ≥ 2 , the BSD conjecture remains open; but the prismatic reformulation is now cleaner: it is a statement about the local operator Φ_p at a single prime, not about a global infinite-dimensional operator whose convergence is itself a conjecture. This is a genuine structural simplification.

5.4. Remaining gap: RH and the Nygaard Extension Conjecture. The one remaining structural gap for the RH reformulation is the passage from the local, associated-graded level (where everything works and N_X is self-adjoint) to the global, full-cohomology level (where Φ_{glob} involves a sum over all primes).

The Nygaard Extension Conjecture reframes this gap as a question in p -adic Hodge theory: are the extension classes $[\varepsilon_r]$ uniformly bounded in p -adic norm? In the ordinary elliptic curve case, the answer is yes (Proposition 3.5). For general motives, the answer depends on the arithmetic of the Nygaard filtration in a way that is currently inaccessible but that does not involve GRH directly.

If the Nygaard Extension Conjecture is true, then:

- $\Phi_{\text{glob}} = N_X + E_{\text{ext}}$ is a bounded perturbation of a self-adjoint operator by a bounded self-adjoint operator;
- by the Kato–Rellich theorem, Φ_{glob} is self-adjoint;
- the convergence of $\Phi_{\text{glob}}^{(P)}$ in operator norm follows from the uniform bound on $\|E_{\text{ext}}\|_{\text{op}} \leq C(X) \cdot p^{-1/2}$.

This would resolve Conjecture 4.1 of Part IV without any appeal to GRH.

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